

Research Paper

Obstacle avoidance guidance for Mars powered descent using convex optimization and elevation angle

Duozhi Gao^a, Youmin Gong^b, Yanning Guo^{a,c,*}, Yizheng Xiao^d,
Edoardo Fadda^e, Paolo Brandimarte^e

^a Department of Control Science and Engineering, Harbin Institute of Technology, Harbin 150001, China

^b School of Intelligence Science and Engineering, Harbin Institute of Technology (Shenzhen), Shenzhen 518055, China

^c Zhengzhou Research Institute, Harbin Institute of Technology, Zhengzhou 450000, China

^d School of Aerospace Science, Harbin Institute of Technology (Shenzhen), Shenzhen 518055, China

^e Department of Mathematical Sciences, DISMA, Politecnico di Torino, Torino 10129, Italy

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ABSTRACT

This paper presents a **convex optimization framework** for obstacle avoidance guidance during the powered descent phase of Mars landing. The proposed approach **retains the computational efficiency of convex optimization while enhancing the onboard camera field of view** at the terminal phase. First, **the relaxed glide-slope constraint and the stepwise constraint** are introduced. The trajectory optimization problem is then mathematically formulated, followed by convexification and discretization. Second, a **homotopy iteration method** is integrated into the sequential convex optimization framework to develop relaxed glide-slope and stepwise obstacle avoidance constraints. Third, leveraging the concept of **elevation angle**, an axis-weighted position integral is introduced as the objective function, followed by the development of an elevation-angle-based convex optimization algorithm. To assess the algorithm's effectiveness, three scenarios are evaluated: fixed fuel and terminal time, fixed terminal time, and fixed fuel. Furthermore, an optimal obstacle avoidance framework is formulated based on **the maximum elevation angle integral**. Simulation results demonstrate that the proposed homotopy iteration-based obstacle avoidance algorithm effectively circumvents obstacles under varying initial conditions while satisfying relaxed glide-slope and stepwise constraints. More importantly, the new convex optimization framework not only enhances obstacle avoidance, but also enables near-vertical landings, thereby improving the terminal camera field of view and reducing the risk of lander tipping upon touchdown. Additionally, simulations confirm that the proposed convex optimization algorithm can achieve effective obstacle avoidance and an enhanced terminal field of view without increasing fuel consumption.

1. Introduction

The rapid advancement of space technology in recent years has greatly increased global interest in planetary surface exploration, particularly targeting celestial bodies such as the Moon, Mars, and asteroids [1]. A series of landmark missions have been successfully executed or proposed, including Martian surface exploration [2], asteroid sampling [3], crewed lunar landings [4], and Mars sample return

initiatives. **Historically, planetary landing missions prioritized safety by selecting flat, obstacle-free regions. However, emerging scientific objectives now demand exploration of complex terrains such as craters, canyons, and volcanic regions, imposing stringent requirements on precision landing and obstacle avoidance capabilities.**

Safe and reliable guidance during the **terminal powered descent phase** becomes particularly critical for achieving **soft pinpoint landings** in these hazardous environments. During this phase, the lander ignites its main engines to generate substantial thrust for deceleration and

* Corresponding author.

E-mail addresses: 21b904021@stu.hit.edu.cn (D. Gao), gongyoumin@hit.edu.cn (Y. Gong), guoyn@hit.edu.cn (Y. Guo), 22b361002@stu.hit.edu.cn (Y. Xiao), edoardo.fadda@polito.it (E. Fadda), paolo.brandimarte@polito.it (P. Brandimarte).

¹ Ph.D. Candidate, Department of Control Science and Engineering, Harbin Institute of Technology.

² Associate Professor, School of Intelligence Science and Engineering, Harbin Institute of Technology (Shenzhen).

³ Professor, Department of Control Science and Engineering, Harbin Institute of Technology.

⁴ Ph.D. Student, School of Aerospace Science, Harbin Institute of Technology (Shenzhen).

⁵ Assistant Professor, Department of Mathematical Sciences, DISMA, Politecnico di Torino.

⁶ Professor, Department of Mathematical Sciences, DISMA, Politecnico di Torino.

trajectory control, ultimately achieving zero terminal velocity at the designated landing site. This process must satisfy numerous internal and external constraints, including thruster magnitude limits and obstacle avoidance requirements. Current guidance methodologies for powered descent can be broadly categorized into three paradigms [5]: analytical guidance [6], numerical optimization-based methods [7], and learning-based approaches [8].

Analytical guidance approaches, such as polynomial guidance [9] and zero-effort miss/zero-effort velocity guidance [10], are widely used for their computational efficiency. Polynomial-based methods have been successfully applied in Apollo and Mars Science Laboratory missions [9], while augmented schemes with tunable parameters have improved their adaptability [11,12]. ZEM/ZEV methods have been extended to incorporate waypoint optimization [13], barrier Lyapunov functions for terrain avoidance [14], and hybrid formulations under relaxed constraints [10,15]. Convex curvature guidance [16–18] combines constant acceleration with ZEM/ZEV to ensure obstacle avoidance. Despite these advances, analytical methods generally struggle with thrust magnitude constraints, parameter sensitivity, and fuel optimality. Learning-based approaches, particularly reinforcement learning [8,19], offer fast online computation but face challenges in stability and reliability for practical deployment.

In contrast, optimization-based guidance methods provide a more direct and effective approach to constraint handling and fuel minimization. In three-degree-of-freedom (3-DoF) scenarios, convex programming and pseudospectral optimization are commonly employed [20, 21], while six-degree-of-freedom (6-DOF) scenarios typically rely on sequential convex optimization and nonlinear programming to address nonlinearity [22,23]. Among these, convex optimization stands out for its rapid computational capabilities. Açıkmüşe et al. [20] first introduced the use of lossless convex optimization techniques for solving the 3-DoF powered descent problem, establishing a basis for later convex programming methods. Building on this work, Blackmore et al. [24] applied similar techniques to address the minimum landing error problem. Scharf et al. [25] developed an onboard software system known as “G-FOLD”, which leverages convex optimization techniques and has demonstrated successful performance in real-world tests. To enhance computational performance, Sagliano [21,26] introduced a hybrid approach combining pseudospectral methods with convex optimization to improve computational efficiency. Song et al. [27] segmented the powered descent phase and addressed the problem using a multi-phase pseudospectral method within a convex optimization framework. Collaborative optimization methods have also been explored to balance fuel efficiency, landing accuracy, and trajectory flexibility [28]. Zhang et al. [29] proposed a parallel sequential convex programming method for cooperative multi-vehicle trajectory replanning. Additionally, Chen et al. [23] utilized a combination of interior-point algorithms and homotopy methods to tackle the 6-DOF landing guidance problem. Optimization-based approaches offer precise constraint handling and enhance fuel efficiency, thereby supporting heavier scientific payloads and increasing the probability of mission success. However, these methods often lead to nonconvex optimization problems, which are computationally intensive and inefficient to solve.

More recently, homotopy iteration has emerged as an effective technique to handle non-convex constraints within optimization-based frameworks. Ma et al. [30] combined the NLP solver with homotopy-based backtracking strategy for lunar rover vertical takeoff and landing maneuvers in the presence of terrain constraints. Wang et al. [31] proposed a homotopic-iterative strategy for rocket landing trajectory optimization with nonlinear aerodynamic drag force. Malyuta and Açıkmüşe [32] introduced a fast homotopy method for spacecraft rendezvous trajectory optimization with discrete logic constraints, significantly improving computational efficiency. Additionally, Zhao et al. [33,34] advanced this methodology by developing smoothing-homotopy-based sequential convex programming for general trajectory optimization problems, with successful applications to ascent trajectory optimization subject to complex path constraints. In a sequence of papers, Pan et al. proposed a family of smoothing homotopy methods

for solving nonlinear optimal control problems, including generalized Gaussian [35], heat-equation-based [36], and Mainardi kernel smoothing approaches [37]. Based on these advances, this paper develops a homotopy-convex optimization method specifically tailored to the challenges of Mars landing, including fuel-optimal guidance with guaranteed obstacle avoidance and an enhanced terminal field of view achieved through elevation-angle maximization.

For obstacle avoidance, Cui et al. [38] integrated convex trajectory design and adaptive curvature regulation within the convex optimization framework to enable obstacle avoidance. Long et al. [39] proposed a vector trajectory-based obstacle avoidance approach that decomposes avoidance constraints into trajectory direction constraint and distance constraint, which are then reformulated as auxiliary angular constraint. Bai et al. [40] employed a relaxation factor and convex relaxation techniques to address conical obstacle constraint protruding from the surface. Liu et al. [41] modeled obstacles using polynomials and incorporated safety distance constraint to maintain a secure margin between the lander and obstacles. Gong et al. introduced a prescribed performance function to address step-shaped terrains [42]. Additionally, methods addressing uncertainty have been developed for safer obstacle avoidance. For example, Yang et al. [43] used collision probability sliding-mode guidance to account for state uncertainties, while Xiao et al. [44] applied polynomial chaos expansion and model predictive convex programming to mitigate navigation errors during landing and ensure safe obstacle avoidance.

Among the various optimization techniques, convex optimization has attracted widespread attention due to its tractability. An ideal scenario for optimization-based obstacle avoidance is to represent the problem through a convex optimization framework. This paper builds upon lossless convexification and proposes an elevation-angle-based convex optimization method. The proposed approach explicitly incorporates relaxed glide-slope constraint and stepwise obstacle avoidance, while ensuring an adequate terminal camera field of view and enabling a controlled landing with minimal lateral velocity to prevent tipping. This work makes the following key contributions:

- (1) A sequential convex optimization framework integrating homotopy iteration is developed to address obstacle avoidance under relaxed glide-slope and stepwise constraints.
- (2) An elevation-angle-based convex optimization algorithm is formulated, incorporating an axis-weighted position integral as the objective function to ensure effective obstacle avoidance while optimizing the terminal field of view.
- (3) A systematic framework for generating trajectories with optimal elevation angle integral is constructed based on the proposed convex optimization algorithm.

The rest of this paper is organized as follows. Section 2 details the problem formulation. Section 3 discusses the improved lossless convexification and obstacle avoidance methods. Section 4 introduces the elevation-angle-based guidance approach. Section 5 provides numerical demonstrations and analysis. Finally, Section 6 concludes the paper.

2. Problem formulation 建模：着陆器看成质点

In this section, a mathematical formulation of the powered descent landing scenario within a 3-DoF framework is presented.

2.1. Mathematical modeling of dynamics

This paper utilizes a powered descent model based on the 3-DoF framework. During the powered descent phase, the lander is modeled as a point mass influenced by gravity and thruster-generated forces. Consistent with most previous studies, this work assumes constant gravitational acceleration, neglects atmospheric drag, and ignores the effects of planetary rotation. The surface inertial coordinate system O_{Ixyz} is defined with its origin located at the targeted landing site. In

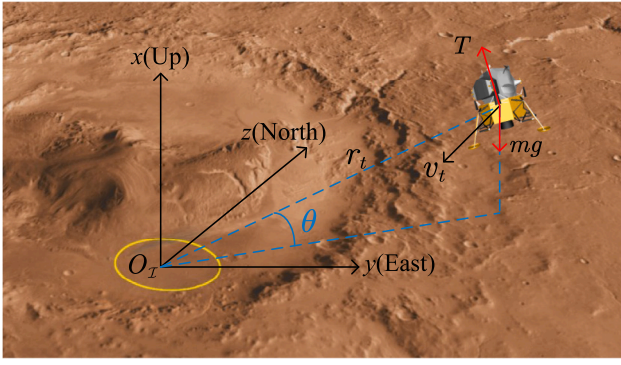


Fig. 1. Coordinate system and powered descent scenario.

this system, the x -axis points vertically upward, the y -axis extends toward the east, and the z -axis points northward. A schematic illustration of this coordinate system is provided in Fig. 1.

Within the surface inertial coordinate system O_Ixyz , the lander's position is expressed as $\mathbf{r} = [r_x, r_y, r_z]^T \in \mathbb{R}^3$, and its velocity as $\mathbf{v} = [v_x, v_y, v_z]^T \in \mathbb{R}^3$. The thrust vector is represented by $\mathbf{T} = [T_x, T_y, T_z]^T \in \mathbb{R}^3$, while the gravitational acceleration on Mars is denoted as $\mathbf{g}_m$. The lander's dynamics are then described by:

$$\begin{aligned} \dot{\mathbf{r}}(t) &= \mathbf{v}(t), \\ \dot{\mathbf{v}}(t) &= \mathbf{g}_m + \frac{\mathbf{T}(t)}{m(t)}, \end{aligned} \quad \text{动力学方程} \quad (1)$$

$$\dot{m}(t) = -\lambda \|\mathbf{T}(t)\|_2.$$

where, m denotes the mass of the lander, λ is the fuel consumption coefficient, $\|\mathbf{T}(t)\|_2$ indicates the magnitude of the thrust. The fuel consumption coefficient λ is calculated using the specific impulse I_{sp} , Earth's gravitational acceleration g_e , and the thruster's configuration, following the approach outlined in [20]:

$$\lambda = \frac{1}{I_{sp} g_e \cos \phi}. \quad (2)$$

An elevation angle, which can be used to characterize the lander's field of view, is represented as θ as shown in Fig. 1. It is defined by

$$\theta \doteq \arcsin \frac{r_x}{\sqrt{r_x^2 + r_y^2 + r_z^2}} = \arctan \frac{r_x}{\sqrt{r_y^2 + r_z^2}}. \quad (3)$$

where, $\sqrt{r_y^2 + r_z^2} \neq 0$.

2.2. Constraint formulation for states and controls

For the soft pinpoint landing problem, the lander's initial and final states are subject to the following constraints:

$$\begin{aligned} \mathbf{r}(t_0) &= \mathbf{r}_0, \quad \mathbf{v}(t_0) = \mathbf{v}_0, \quad m(t_0) = m_{\text{wet}}, \\ \mathbf{r}(t_f) &= \mathbf{r}_f, \quad \mathbf{v}(t_f) = \mathbf{v}_f, \quad m(t_f) \geq m_{\text{dry}}. \end{aligned} \quad (4)$$

where t_0 and t_f denote the initial and final times, respectively. m_{wet} represents the total initial mass (lander plus fuel), while m_{dry} is the dry mass of the lander. The mass values at the start and end of the maneuver are $m(t_0)$ and $m(t_f)$, respectively. Similarly, $\mathbf{r}_0$ and $\mathbf{r}_f$ correspond to the initial and final positions, and $\mathbf{v}_0$ and $\mathbf{v}_f$ represent the initial and final velocities.

Control constraints arise from the physical characteristics and assumptions of the thrusters. The constraints on thrust are as follows:

$$0 < T_{\min} \leq \|\mathbf{T}(t)\|_2 \leq T_{\max}. \quad (5)$$

where $T_{\min}$ is a positive value, indicating that the engines cannot be shut down during the descent process and must be started from the beginning, while $T_{\max}$ represents the maximum thrust the thrusters can provide.

2.3. Mathematical modeling of obstacle avoidance

Obstacle avoidance is critical during the landing process. Some existing studies model obstacles using a **glide-slope constraint** to ensure the lander does not collide with them. In this approach, the glide-slope constraint increases continuously with altitude. However, during descent, when the altitude exceeds a certain safety threshold, the risk of collision with obstacles becomes negligible. To address this, this study enhances the conventional glide-slope constraint by introducing a relaxed glide-slope constraint or a stepwise constraint, as shown in Fig. 2. The conventional glide-slope constraint, illustrated in Fig. 2(a), is mathematically expressed as follows:

$$\tan \gamma_{gs} \leq \frac{r_x}{\sqrt{r_y^2 + r_z^2}}. \quad \text{7月26日位置} \quad (6)$$

where, γ_{gs} denotes the glide-slope angle, r_x , r_y , and r_z are the individual components of the position vector $\mathbf{r}$.

The relaxed glide-slope constraint, as illustrated in Fig. 2(b), is mathematically defined as follows:

$$r_x \geq \begin{cases} 0, & \sqrt{r_y^2 + r_z^2} \leq l_1 \\ (\sqrt{r_y^2 + r_z^2} - l_1) \tan \gamma_{\text{new}}, & l_1 < \sqrt{r_y^2 + r_z^2} < l_2 \\ h, & l_2 \leq \sqrt{r_y^2 + r_z^2} \end{cases} \quad (7)$$

where, $\tan \gamma_{\text{new}} = h / (l_2 - l_1)$ denotes the relaxed glide-slope angle, h represents the safe altitude. l_1 is the safe radius at the landing site, and l_2 is the safe radius at an altitude of h .

The stepwise constraint, as illustrated in Fig. 2(c), is mathematically defined as follows:

$$r_x \geq \begin{cases} 0, & \sqrt{r_y^2 + r_z^2} \leq l \\ h, & l \leq \sqrt{r_y^2 + r_z^2} \end{cases} \quad (8)$$

where, h is the safe altitude, and l is the safe radius.

2.4. Fuel optimal problem formulation

Fuel consumption is one of the important metrics in the landing process, especially for extraterrestrial objects such as the Moon and Mars. Under these circumstances, the optimization problem aims to minimize overall fuel usage. The thrust magnitude and direction serve as the optimization variables. The fuel-optimal powered descent guidance task is posed as the following optimal control problem:

Problem 1. Fuel optimal powered descent problem

$$\begin{aligned} \min_{\mathbf{T}(t)} -m(t_f) &\Leftrightarrow \min_{\mathbf{T}(t)} \int_0^{t_f} \|\mathbf{T}(t)\|_2 dt \\ \text{subject to :} & \text{Eqs. (1), (4), (5),} \\ & \text{Eq. (6) or (7) or (8).} \end{aligned} \quad (9)$$

Problem 1 describes the fuel optimal powered descent guidance problem, which is inherently non-convex in nature. When employing conventional glide-slope constraint in Eq. (6) for obstacle avoidance, an effective approach to addressing this challenge involves the utilization of lossless convexification to reformulate the original non-convex problem as a second-order cone programming (SOCP) problem [20,24]. When incorporating the relaxed glide-slope constraint in Eq. (7) or the stepwise constraint in Eq. (8), these constraints cannot be directly formulated as second-order cone constraints.

3. Improved lossless convexification and obstacle avoidance

Building upon the standard lossless convexification framework [20, 24] and homotopy-based obstacle handling, this section presents a

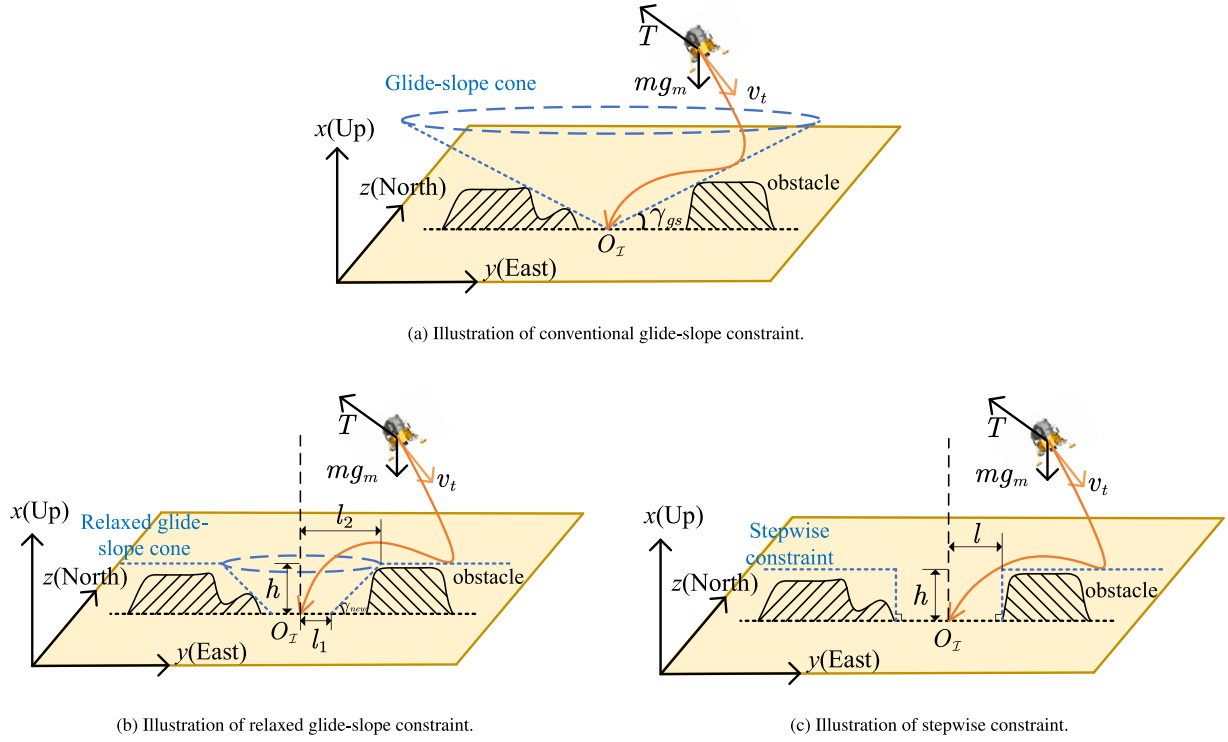


Fig. 2. Illustration of obstacle avoidance constraints.

discrete-time convex formulation of the powered descent problem. Compared with the conventional formulation, the proposed method introduces an iterative refinement scheme to reduce Taylor approximation errors and incorporates obstacle avoidance through homotopy iteration and slack relaxation.

3.1. Lossless convexification

For **Problem 1**, the non-convexity arises primarily from the minimum thrust constraint and the nonlinear dynamics involving mass division. Following Ref. [20], a slack variable $\Gamma(t)$ is introduced to relax the thrust constraint:

$$\begin{aligned} \dot{\mathbf{r}}(t) &= \mathbf{v}(t), \quad \dot{\mathbf{v}}(t) = \mathbf{g}_m + \frac{\mathbf{T}(t)}{m(t)}, \\ \dot{m}(t) &= -\lambda \Gamma(t), \quad \|\mathbf{T}(t)\|_2 \leq \Gamma(t), \\ 0 < T_{\min} &\leq \Gamma(t) \leq T_{\max}. \end{aligned} \quad (10)$$

where $\Gamma(t)$ is the introduced slack variable. As shown in Refs. [20,24], this relaxation is lossless under fuel-optimal objectives.

To eliminate nonlinearities in dynamics, the following substitutions are introduced:

$$z(t) \doteq \ln m(t), \quad \mathbf{u}(t) \doteq \frac{\mathbf{T}(t)}{m(t)}, \quad \sigma(t) \doteq \frac{\Gamma(t)}{m(t)}. \quad (11)$$

The dynamics, initial and final state constraints are rewritten as follows:

$$\begin{aligned} \dot{\mathbf{r}}(t) &= \mathbf{v}(t), \quad \dot{\mathbf{v}}(t) = \mathbf{g}_m + \mathbf{u}(t), \\ \dot{z}(t) &= -\lambda \sigma(t), \quad \|\mathbf{u}(t)\|_2 \leq \sigma(t), \\ \mathbf{r}(t_0) &= \mathbf{r}_0, \mathbf{v}(t_0) = \mathbf{v}_0, z(t_0) = \ln m_{\text{wet}}, \\ \mathbf{r}(t_f) &= \mathbf{r}_f, \mathbf{v}(t_f) = \mathbf{v}_f, z(t_f) \geq \ln m_{\text{dry}}. \end{aligned} \quad (12)$$

The trust constraint is restated as follows:

$$0 < \frac{T_{\min}}{m(t)} \leq \sigma(t) \leq \frac{T_{\max}}{m(t)}. \quad (13)$$

Under this transformation, the fuel-optimal objective becomes

$$\min_{\mathbf{u}(t), \sigma(t)} \int_0^{t_f} \sigma(t) dt. \quad (14)$$

Consequently, **Problem 1** can be transformed into the following relaxed formulation:

Problem 2. Relaxed fuel optimal powered descent problem

$$\min_{\mathbf{u}(t), \sigma(t)} -z(t_f) \Leftrightarrow \min_{\mathbf{u}(t), \sigma(t)} \int_0^{t_f} \sigma(t) dt \quad (15)$$

subject to : Eqs. (6), (12), (13).

The thrust bound becomes

$$0 < T_{\min} e^{-z(t)} \leq \sigma(t) \leq T_{\max} e^{-z(t)}. \quad (16)$$

which is non-convex. A Taylor approximation is applied to obtain a convex relaxation:

$$T_{\min} e^{-z_l(t)} \left[1 - [z(t) - z_l(t)] + \frac{1}{2} [z(t) - z_l(t)]^2 \right] \leq \sigma(t), \quad (17)$$

$$\sigma(t) \leq T_{\max} e^{-z_u(t)} \left[1 - [z(t) - z_u(t)] \right].$$

where, $z_l(t)$ and $z_u(t)$ are defined as:

$$\begin{aligned} z_l(t) &= \ln(m_{\text{wet}} - \lambda T_{\min} t), \\ z_u(t) &= \ln(m_{\text{wet}} - \lambda T_{\max} t). \end{aligned} \quad (18)$$

In general, **Problem 2** can be reformulated through variable substitution and Taylor expansion as follows:

Problem 3. Convexified fuel optimal powered descent problem

$$\min_{\mathbf{u}(t), \sigma(t)} -z(t_f) \Leftrightarrow \min_{\mathbf{u}(t), \sigma(t)} \int_0^{t_f} \sigma(t) dt \quad (19)$$

subject to : Eqs. (6), (12), (17), (18).

3.2. Discretization

Problem 3 constitutes a convex optimization task, specifically a SOCP problem. To efficiently solve this problem using convex optimization solvers, it is necessary to discretize the continuous formulation. By

defining a new state variable vector $\mathbf{X}(t)$, the discrete system can be expressed as follows:

$$\dot{\mathbf{X}}(t) = \mathbf{A}_c \mathbf{X}(t) + \mathbf{B}_c \begin{bmatrix} \mathbf{u}(t) \\ \sigma(t) \end{bmatrix} + \mathbf{B}_c \mathcal{G}_m. \quad (20)$$

where,

$$\mathbf{X}(t) = \begin{bmatrix} \mathbf{r}(t) \\ \mathbf{v}(t) \\ z(t) \end{bmatrix}, \mathcal{G}_m = \begin{bmatrix} \mathbf{g}_m \\ 0 \end{bmatrix}, \quad (21)$$

$$\mathbf{A}_c = \begin{bmatrix} 0_{3 \times 3} & \mathbf{I}_3 & 0_{3 \times 1} \\ 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 1} \\ 0_{1 \times 3} & 0_{1 \times 3} & 0_{1 \times 1} \end{bmatrix}, \mathbf{B}_c = \begin{bmatrix} 0_{3 \times 3} & 0_{3 \times 1} \\ \mathbf{I}_3 & 0_{3 \times 1} \\ 0_{1 \times 3} & -\lambda \end{bmatrix}.$$

The system is discretized through a zero-order hold (ZOH) approach, where the sampling interval Δt is selected as follows:

$$\Delta t = \frac{t_f - t_0}{N + 1}. \quad (22)$$

Let $\mathbf{X}(t_k)$ denote the discrete state variable at time t_k , and let $\mathbf{u}(t_k)$ and $\sigma(t_k)$ represent the discrete control variables. The discretized system equations are expressed as follows:

$$\mathbf{X}(t_{k+1}) = \mathbf{A}_d \mathbf{X}(t_k) + \mathbf{B}_d \begin{bmatrix} \mathbf{u}(t_k) \\ \sigma(t_k) \end{bmatrix} + \mathbf{B}_d \mathcal{G}_m, \quad k = 0, 1, \dots, N. \quad (23)$$

where, the matrices $\mathbf{A}_d$ and $\mathbf{B}_d$ are computed as follows:

$$\mathbf{A}_d = e^{\mathbf{A}_c \Delta t}, \mathbf{B}_d = \int_0^{\Delta t} e^{\mathbf{A}_c(\Delta t - \tau)} \mathbf{B}_c d\tau. \quad (24)$$

The thrust constraint, initial and terminal state constraints in Eq. (12) can be converted into a discretized form as follows:

$$\|\mathbf{u}(t_k)\|_2 \leq \sigma(t_k), \mathbf{X}(t_0) = \begin{bmatrix} \mathbf{r}_0 \\ \mathbf{v}_0 \\ \ln m_{\text{wet}} \end{bmatrix}, \quad (25)$$

$$\begin{bmatrix} \mathbf{I}_3 & 0_{3 \times 3} & 0_{3 \times 1} \\ 0_{3 \times 3} & \mathbf{I}_3 & 0_{3 \times 1} \\ 0_{3 \times 1} & 0_{3 \times 1} & 1 \end{bmatrix} \mathbf{X}(t_f) = \begin{bmatrix} \mathbf{r}_f \\ \mathbf{v}_f \end{bmatrix},$$

$$[0_{3 \times 1} \quad 0_{3 \times 1} \quad 1] \mathbf{X}(t_f) \geq \ln m_{\text{dry}}.$$

The conventional glide-slope constraint in (6) and linearized constraints in (17) are described as the following discrete form:

$$T_{\min} e^{-z_l(t_k)} \left[1 - [z(t_k) - z_l(t_k)] + \frac{1}{2} [z(t_k) - z_l(t_k)]^2 \right] \leq \sigma(t_k),$$

$$\sigma(t_k) \leq T_{\max} e^{-z_u(t_k)} [1 - [z(t_k) - z_u(t_k)]]], \quad (26)$$

$$\sqrt{r_{y,k}^2 + r_{z,k}^2} \tan \gamma_{gs} \leq r_{x,k}.$$

where, $k = 0, 1, \dots, N$. The intermediate variables are calculated as follows:

$$t_k = \Delta t \cdot k = \frac{t_f - t_0}{N + 1} k,$$

$$z_l(t_k) = \ln(m_{\text{wet}} - \lambda T_{\min} t_k),$$

$$z_u(t_k) = \ln(m_{\text{wet}} - \lambda T_{\max} t_k), \quad (27)$$

$$r_{x,k} = [1 \quad 0_{1 \times 6}] \mathbf{X}(t_k), r_{y,k} = [0 \quad 1 \quad 0_{1 \times 5}] \mathbf{X}(t_k),$$

$$r_{z,k} = [0 \quad 0 \quad 1 \quad 0_{1 \times 4}] \mathbf{X}(t_k).$$

Thus, the discretized formulation of the convexified fuel optimal powered descent Problem 3 is as follows:

Problem 4. Discretized and convexified fuel optimal powered descent problem

$$\min_{\mathbf{u}(t_k), \sigma(t_k)} -z(t_{N+1}) \Leftrightarrow \min_{\mathbf{u}(t_k), \sigma(t_k)} \lambda \Delta t \sum_{k=0}^N \sigma(t_k) \quad (28)$$

subject to : Eqs. (23), (25), (26), (27).

3.3. Improved lossless convexification

Problem 4 presents a discrete model that can be directly solved using convex optimization solvers, enabling convenient modeling and solution implementation within frameworks such as CVX or YALMIP. However, this model relies on a Taylor expansion to achieve a convex approximation of the inherently non-convex problem, leading to approximation errors that accumulate over time t . To address this limitation and achieve a more accurate solution to the original problem, an iterative refinement algorithm based on the solution of Problem 4 is proposed.

Let $\mathbf{u}(\cdot)$ represent the set $\{\mathbf{u}(t_k) \mid k = 0, 1, \dots, N\}$, $\sigma(\cdot)$ denote the set $\{\sigma(t_k) \mid k = 0, 1, \dots, N\}$, and $\mathbf{X}(\cdot)$ signify the set $\{\mathbf{X}(t_k) \mid k = 0, 1, \dots, N + 1\}$. The solution to Problem 4 is denoted as $\tilde{\mathbf{u}}(\cdot)$, $\tilde{\sigma}(\cdot)$, and $\tilde{\mathbf{X}}(\cdot)$. Accordingly, $\tilde{z}(t_k)$ is given by:

$$\tilde{z}(t_k) = [0_{1 \times 6} \quad 1] \tilde{\mathbf{X}}(t_k). \quad (29)$$

Therefore, based on Eq. (29), the original approximation in Eq. (26) is transformed into a modified Taylor expansion around $\tilde{z}(t_k)$, as follows:

$$T_{\min} e^{-\tilde{z}(t_k)} \left[1 - [z(t_k) - \tilde{z}(t_k)] + \frac{1}{2} [z(t_k) - \tilde{z}(t_k)]^2 \right] \leq \sigma(t_k), \quad (30)$$

$$\sigma(t_k) \leq T_{\max} e^{-\tilde{z}(t_k)} [1 - [z(t_k) - \tilde{z}(t_k)]]].$$

3.4. Obstacle avoidance using homotopy iteration

To generalize the convex constraints in Eq. (6) to the non-convex segmented forms in Eqs. (7) and (8), this work employs a homotopy iteration method. By gradually introducing stricter obstacle constraints to refine collision avoidance, the approach ensures smooth computation and generates safe obstacle avoidance trajectories. The core idea is to construct a continuous deformation process that transitions from a simple convex-constrained path to a complex non-convex-constrained path.

First, given a reference trajectory $\mathbf{X}(\cdot)$, Eqs. (6), (7) and (8) are unified and expressed as follows:

$$r_x(t_k) \geq F(\mathbf{X}(t_k), \delta). \quad (31)$$

where, $0 \leq \delta \leq 1$. As for relaxed glide-slope constraint (7), $F(\mathbf{X}(t_k), \delta)$ is calculated as in Box I.

For stepwise constraint (8), $F(\mathbf{X}(t_k), \delta)$ is calculated as:

$$F(\mathbf{X}(t_k), \delta) = \begin{cases} 0, & \sqrt{r_y(t_k)^2 + r_z(t_k)^2} \leq l \\ \delta \cdot h, & l \leq \sqrt{r_y(t_k)^2 + r_z(t_k)^2} \end{cases} \quad (32)$$

Next, since the obstacle constraints are derived based on the reference trajectory and, in addition to these constraints, thrust magnitude constraint also exist, directly incorporating Eq. (31) as a constraint may render the problem unsolvable. To address this, a slack variable is introduced, and Eq. (31) is further reformulated as follows:

$$r_x(t_k) \geq F(\mathbf{X}(t_k), \delta) + \zeta(t_k). \quad (34)$$

where ζ is the slack variable. To ensure that ζ decreases iteratively, the corresponding objective function is formulated as follows:

$$\min_{\mathbf{u}(t_k), \sigma(t_k)} -z(t_{N+1}) + W_\zeta \cdot \|\zeta(\cdot)\|_2 \Leftrightarrow \min_{\mathbf{u}(t_k), \sigma(t_k)} \lambda \Delta t \sum_{k=0}^N \sigma(t_k) + W_\zeta \cdot \|\zeta(\cdot)\|_2. \quad (35)$$

Additionally, to generate the initial trajectory, the conventional glide-slope angle is set to $\gamma_{gs} = 0$, and Problem 4 is solved. The resulting solution, denoted as $\tilde{\mathbf{u}}(\cdot)$, $\tilde{\sigma}(\cdot)$, and $\tilde{\mathbf{X}}(\cdot)$, forms the foundation for constructing the obstacle avoidance problem as follows:

$$F(X(t_k), \delta) = \begin{cases} \delta \cdot 0, & \sqrt{r_y(t_k)^2 + r_z(t_k)^2} \leq l_1 \\ \delta \cdot \left(\sqrt{r_y(t_k)^2 + r_z(t_k)^2} - l_1 \right) \tan \gamma_{new}, & l_1 < \sqrt{r_y(t_k)^2 + r_z(t_k)^2} < l_2 \\ \delta \cdot h, & l_2 \leq \sqrt{r_y(t_k)^2 + r_z(t_k)^2} \end{cases} \quad (32)$$

Box I.

Problem 5. Discrete fuel optimal powered descent with obstacle avoidance

$$\min_{u(t_k), \sigma(t_k)} -z(t_{N+1}) + W_\zeta \cdot \|\zeta(\cdot)\|_2 \Leftrightarrow \min_{u(t_k), \sigma(t_k)} \lambda \Delta t \sum_{k=0}^N \sigma(t_k) + W_\zeta \cdot \|\zeta(\cdot)\|_2$$

subject to : Eqs. (23), (25), (30), (34)
Eqs. (32) or (33).

(36)

Building on the previous discussion, the procedure for the *Improved lossless convexification algorithm with obstacle avoidance for fuel optimal powered descent* is outlined in **Algorithm 1** as follows:

Algorithm 1 Improved lossless convexification algorithm with obstacle avoidance for fuel optimal powered descent.

- 1: Define tolerance $\varepsilon_\zeta > 0$. Define homotopy iteration count $H \in \mathbb{Z}^+$.
- 2: Define iteration count $j = 0$. Define Endflag=false.
- 3: Initial trajectory: solve **Problem 4** with $\gamma_{gs} = 0$ to obtain $\hat{u}(\cdot), \hat{\sigma}(\cdot), \hat{X}(\cdot)$.
- 4: **while** Endflag=false **do**
- 5: $j = j + 1$.
- 6: Assign $\delta = \min(j, H) / H$.
- 7: Calculate $F(\hat{X}(t_k), \delta)$ according to Eq. (32) or (33).
- 8: Assign $z_l(t_k) = \bar{z}(t_k), z_u(t_k) = \bar{z}(t_k)$ according to Eq. (30).
- 9: Solve **Problem 5**. Denote the solution as $\hat{u}(\cdot), \hat{\sigma}(\cdot), \hat{X}(\cdot)$.
- 10: Compute the norm of the slack variable $\|\zeta(\cdot)\|_2$.
- 11: Check whether the obtained solution satisfies all constraints.
- 12: **if** All constraints satisfied **and** $\|\zeta(\cdot)\|_2 \leq \varepsilon_\zeta$ **and** $j \geq (H+1)$ **then**
- 13: Endflag=true.
- 14: **else**
- 15: Set the solution $\hat{u}(\cdot), \hat{\sigma}(\cdot), \hat{X}(\cdot)$ as the initial trajectory. Assign $\hat{u}(\cdot) = \hat{u}(\cdot), \hat{\sigma}(\cdot) = \hat{\sigma}(\cdot), \hat{X}(\cdot) = \hat{X}(\cdot)$.
- 16: Endflag=false.
- 17: **end if**
- 18: **end while**
- 19: Assign $u^*(\cdot) = \hat{u}(\cdot), \sigma^*(\cdot) = \hat{\sigma}(\cdot), X^*(\cdot) = \hat{X}(\cdot)$.
- 20: **return** $u^*(\cdot), \sigma^*(\cdot), X^*(\cdot)$ as the fuel optimal solution with obstacle avoidance.

Algorithm 1 presents a fuel optimal trajectory optimization method under relaxed glide-slope or stepwise constraint. A homotopy iteration approach is employed to avoid the issue of gradient explosion caused by first-order derivative calculations in sequential convex optimization.

4. Elevation-angle-based powered descent guidance

In this section, the maximization of the elevation angle in Eq. (3) is indirectly incorporated into the convex optimization problem. A new convex optimization framework is proposed, employing an axis-weighted position integral as the objective function. This formulation also enhances obstacle avoidance performance and improves the terminal field of view.

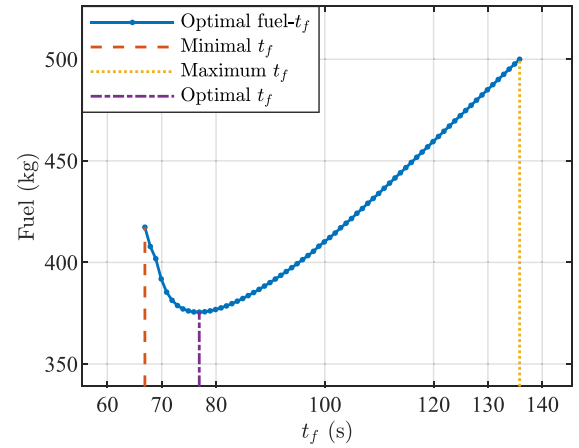


Fig. 3. Relationship between optimal fuel consumption and terminal time t_f .

4.1. Elevation-angle-based convex optimization framework

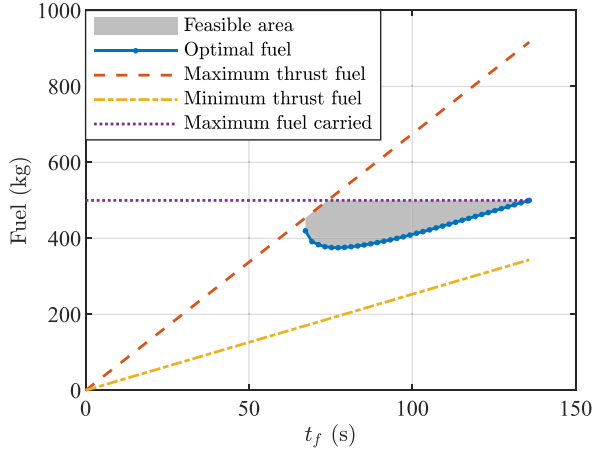
By employing Algorithm 1, the optimal fuel consumption for a given terminal time t_f can be obtained. Given the initial and final state conditions, the minimum and maximum landing times can be determined using tools such as GPOPS software or sequential convex optimization [22]. Alternatively, these terminal times can also be computed by iteratively executing Algorithm 1 until one of the subproblems becomes infeasible. Based on Algorithm 1 and parameters in Tables 1 and 2, the optimal fuel consumption under stepwise constraint is iteratively computed from the minimum to the maximum terminal time. The relationship between optimal fuel consumption and terminal time t_f is presented in Fig. 3.

Fig. 3 illustrates the optimal fuel consumption curve. Taking into account the thrust magnitude constraint, the maximum fuel consumption curve is obtained when the thrust is maintained at its maximum level, while the minimum fuel consumption curve corresponds to the minimum thrust. Given the initial state, which specifies the amount of fuel carried, the feasible region can be derived as shown in Fig. 4.

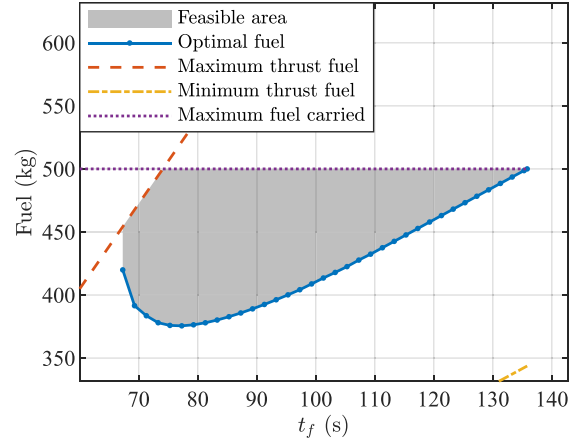
Conventional fuel optimal convex optimization algorithms [20,24], as well as the proposed Algorithm 1, primarily focus on the lower bound of the feasible region. However, in practical scenarios, most cases lie within the feasible region. Therefore, investigating more reasonable obstacle avoidance strategies within this feasible region is of paramount importance.

A straightforward approach is to optimize the use of additional fuel beyond the fuel optimal baseline to improve obstacle avoidance and enhance the terminal field of view. In this context, leveraging the definition of the elevation angle in Eq. (3), the goal is to maximize altitude while minimizing horizontal distance, thereby achieving a larger elevation angle. This strategy enhances both obstacle avoidance and onboard field of view, as illustrated in Fig. 5.

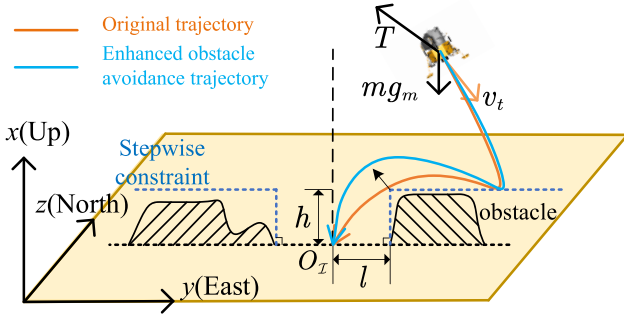
Based on this idea, a new convex optimization framework is proposed with an axis-weighted position integral as the objective function:



(a) Global diagram of feasible area.



(b) Localized enlargement of feasible region.

Fig. 4. Illustration of feasible area for fuel consumption and terminal time t_f .**Fig. 5.** Enhanced obstacle avoidance.

$$\min_{u(t), \sigma(t)} \int_{t_0}^{t_f} (-\alpha \cdot r_x(t) + |r_y(t)| + |r_z(t)|) dt. \quad (37)$$

where $\alpha > 0$ is a tuning coefficient that regulates the extent to which the trajectory is biased toward the vertical direction. However, due to fuel constraints, an excessively large value of α may lead the optimized trajectory to exhibit an undesirable increase in altitude during the initial phase of landing, while limiting altitude adjustment in the terminal phase. Therefore, selecting an appropriate range for α is essential, and this will be discussed in detail in Section 4.2

The discrete form of the objective function (37) is

$$\min_{u(t_k), \sigma(t_k)} -\alpha \cdot \sum_{k=1}^N r_x(t_k) + \sum_{k=1}^N |r_y(t_k)| + \sum_{k=1}^N |r_z(t_k)|. \quad (38)$$

Adopting the objective function as Eq. (38), Problem 4 is converted to the following form:

Problem 6. Elevation-angle-based powered descent problem with basic obstacle avoidance

$$\min_{u(t_k), \sigma(t_k)} -\alpha \cdot \sum_{k=1}^N r_x(t_k) + \sum_{k=1}^N |r_y(t_k)| + \sum_{k=1}^N |r_z(t_k)| \quad (39)$$

subject to : Eqs. (23), (25), (26), (27).

Problem 6 employs conventional glide-slope constraint for obstacle avoidance. To resolve the non-convex constraints in Eqs. (32) and (33) within Problem 5, the discrete form of the improved obstacle avoidance powered descent problem is formulated as follows:

Problem 7. Elevation-angle-based powered descent problem with improved obstacle avoidance

$$\begin{aligned} \min_{u(t_k), \sigma(t_k)} & -\alpha \cdot \sum_{k=1}^N r_x(t_k) + \sum_{k=1}^N |r_y(t_k)| + \sum_{k=1}^N |r_z(t_k)| + W_\zeta \cdot \|\zeta(\cdot)\|_2 \\ \text{subject to :} & \text{Eqs. (23), (25), (30), (34)} \\ & \text{Eqs. (32) or (33).} \end{aligned} \quad (40)$$

where, it is worth noting that the calculation of m_{dry} in Problems 6 and 7 is based on the initial mass and given fuel consumption, expressed as $m_{\text{dry}} = m_{\text{wet}} - m_{\text{fuel}}$.

Since the objective function of Problem 7 is convex, Problem 7 is still a convex optimization problem. When the obstacle avoidance constraint is conventional glide-slope constraint as shown in Eq. (6), Problem 7 can be solved through a single optimization step. However, when the constraint is complex, as defined by Eqs. (32) or (33), a homotopy iteration approach is employed to address the problem.

Building on Problems 6 and 7, An elevation-angle-based convex optimization algorithm with enhanced obstacle avoidance for powered descent guidance is outlined in Algorithm 2 as follows:

Algorithm 2 Elevation-angle-based convex optimization algorithm with enhanced obstacle avoidance for powered descent guidance.

- 1: Define tolerance $\epsilon_\zeta > 0$. Define homotopy iteration count $H \in \mathbb{Z}^+$.
- 2: Define iteration count $j = 0$. Define Endflag=false.
- 3: Using Algorithm 1 to calculate the feasible area as shown in Fig. 4.
- 4: Chose the fuel mass m_{fuel} and terminal time t_f in the feasible area.
- 5: Initial trajectory: solve Problem 6 with selected m_{fuel}, t_f and $\gamma_{\text{gs}} = 0$ to obtain $\hat{u}(\cdot), \hat{\sigma}(\cdot), \hat{X}(\cdot)$.
- 6: **while** Endflag=false **do**
- 7: $j = j + 1$. Assign $\delta = \min(j, H) / H$.
- 8: Calculate $F(\hat{X}(t_k), \delta)$ according to Eq. (32) or (33).
- 9: Assign $z_l(t_k) = \hat{z}(t_k), z_u(t_k) = \hat{z}(t_k)$ according to Eq. (30).
- 10: Solve Problem 7. Denote the solution as $\hat{u}(\cdot), \hat{\sigma}(\cdot), \hat{X}(\cdot)$.
- 11: Compute the norm of the slack variable $\|\zeta(\cdot)\|_2$.
- 12: Check whether the obtained solution satisfies all constraints.
- 13: **if** All constraints satisfied **and** $\|\zeta(\cdot)\|_2 \leq \epsilon_\zeta$ **and** $j \geq (H + 1)$ **then**
- 14: Endflag=true.
- 15: **else**
- 16: Set the solution $\hat{u}(\cdot), \hat{\sigma}(\cdot), \hat{X}(\cdot)$ as the initial trajectory. Assign $\hat{u}(\cdot) = \hat{u}(\cdot), \hat{\sigma}(\cdot) = \hat{\sigma}(\cdot), \hat{X}(\cdot) = \hat{X}(\cdot)$.
- 17: Endflag=false.
- 18: **end if**

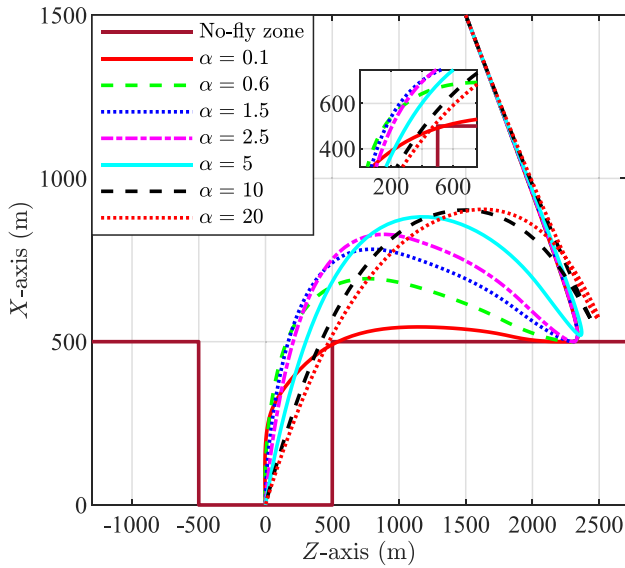


Fig. 6. Enhanced obstacle avoidance trajectories corresponding to different alpha.

19: **end while**
 20: Assign $u^*(\cdot) = \hat{u}(\cdot)$, $\sigma^*(\cdot) = \hat{\sigma}(\cdot)$, $X^*(\cdot) = \hat{X}(\cdot)$.
 21: **return** $u^*(\cdot)$, $\sigma^*(\cdot)$, $X^*(\cdot)$ as the fuel optimal solution with enhanced obstacle avoidance.

Algorithm 2 provides an enhanced obstacle avoidance method applicable to any point within the feasible region illustrated in Fig. 4. The following section analyzes the advantages of this method and the impact of parameters within the algorithm.

4.2. The effect of the coefficient α on the trajectory

The parameter α in the objective function Eq. (38) governs the extent to which the overall trajectory is biased toward the vertical upward direction. A systematic analysis of how varying α influences the trajectory, together with the selection of an appropriate value based on well-defined performance metrics, is therefore of critical importance.

Using the landing parameters provided in Tables 1 and 2, with a fuel consumption of 400 kg. The values $\alpha = 0.1, 0.6, 1.5, 2.5, 5, 10$, and 20 are selected. By applying Algorithm 2, solutions under stepwise constraints are obtained, and the resulting obstacle avoidance trajectories are illustrated in Fig. 6.

According to Fig. 6, it is evident that all trajectories achieve obstacle avoidance. When α is small, such as $\alpha = 0.1$, the overall trajectory exhibits minimal upward deviation, resulting in the trajectory being close to the obstacle at the terminal phase. As α gradually increases, the overall trajectory shifts upward, moving farther away from the obstacle. However, when α is large, such as $\alpha = 10$ or 20, the trajectory's significant upward deviation causes the terminal phase to remain close to the obstacle. Thus, both very small and very large values of α are inappropriate. To more effectively evaluate the obstacle avoidance performance, the elevation angles corresponding to various α values are illustrated in Fig. 7.

As shown in Fig. 7, when α is small, such as $\alpha = 0.1$, the elevation angle at the terminal phase is relatively large, but the elevation angle during the obstacle avoidance segment is smaller compared to other trajectories. Conversely, when α is large, such as $\alpha = 10$ or 20, the terminal elevation angle is significantly smaller than that of other trajectories. Therefore, this study adopts the integral of the elevation angle as a metric to evaluate suitable value of α .

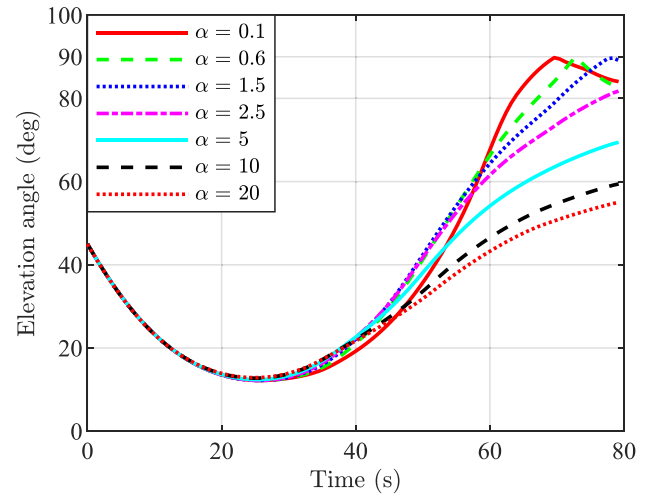


Fig. 7. Elevation angles corresponding to different alpha.

To account for variations in terminal time, the integral is performed with respect to altitude rather than time, as expressed in the following formula:

$$\begin{aligned} \Psi &= \int_{r_{x,f}}^{r_{x,0}} \theta dr_x \\ &= \int_{r_{x,0}}^{r_{x,f}} \arcsin \frac{r_x(t)}{\sqrt{r_x(t)^2 + r_y(t)^2 + r_z(t)^2}} d(-r_x) \\ &= \int_{t_0}^{t_f} \arcsin \frac{r_x(t)}{\sqrt{r_x(t)^2 + r_y(t)^2 + r_z(t)^2}} \cdot (-v_x(t)) dt. \end{aligned} \quad (41)$$

where, the use of the arcsine function helps to resolve ambiguities. To facilitate computation, the integral with respect to altitude is then transformed into an equivalent time-based integral.

The discrete form of the integral of the elevation angle is expressed as

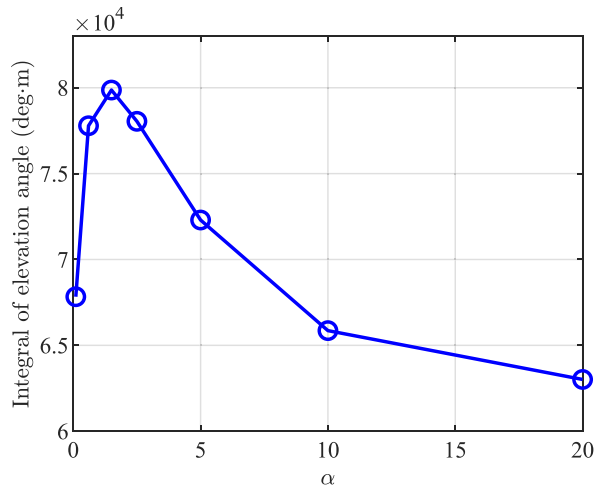
$$\begin{aligned} \Psi &= \sum_{k=1}^N \theta_k \cdot (-v_x(t_k)) \Delta t, \\ \theta_k &= \arcsin \frac{r_x(t_k)}{\sqrt{r_x(t_k)^2 + r_y(t_k)^2 + r_z(t_k)^2}}. \end{aligned} \quad (42)$$

The integrals of the elevation angle corresponding to different values of α in Figs. 6 and 7 are presented in Fig. 8. As shown in the figure, varying α leads to different values of the elevation angle integral. Nevertheless, there exists a range of α that yields relatively large integrals.

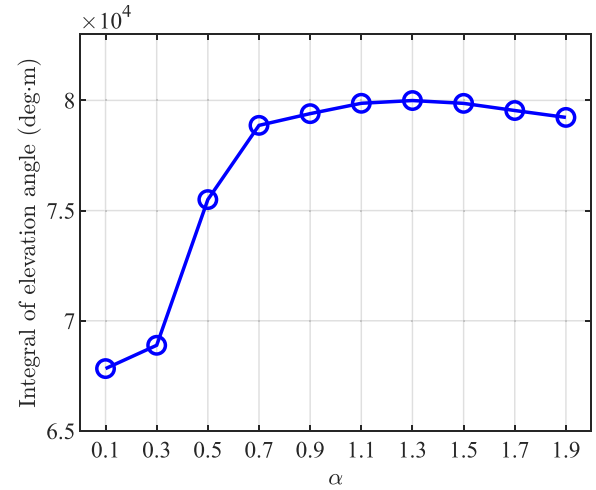
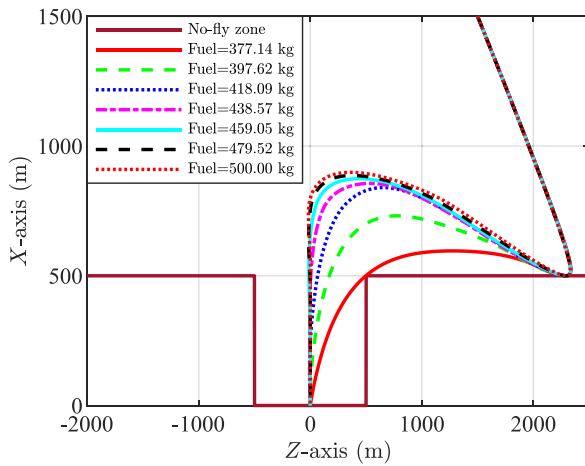
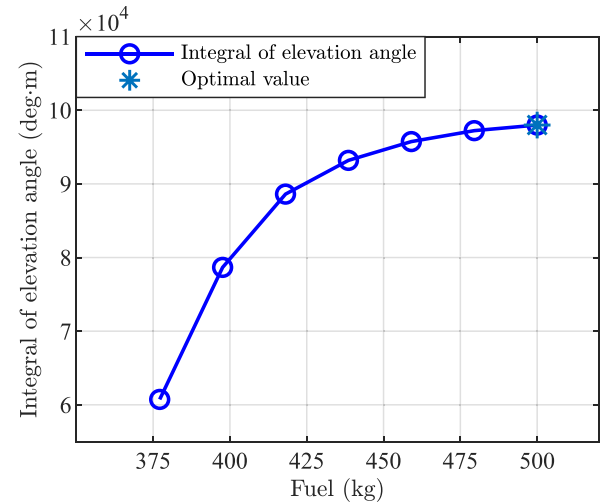
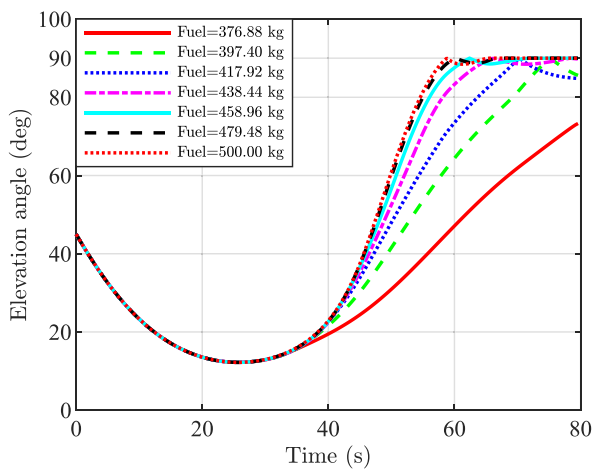
For a fixed initial state, the value of α that maximizes the elevation angle integral can be selected. In cases with non-fixed initial conditions, such as varying terminal times or fuel constraints, a reasonable range of α can instead be identified. Based on extensive simulations, the interval $\alpha \in [0.5, 2]$ is found to be appropriate. When $\alpha = 1$, the contributions of the three axes are equally weighted, it is well-suited for most initial conditions and is therefore set as the default in this paper.

4.3. Trajectory analysis under fixed terminal time and varying fuel consumption

Within the feasible region depicted in Fig. 4, when the terminal time is fixed, the fuel consumption has both an upper and a lower bound. This subsection analyzes how increased fuel consumption affects the elevation angle under a fixed terminal time. Specifically, when the terminal time t_f is set to 80 s and other parameter conditions are as listed in Table 1 and 2, the resulting trajectories and elevation angles are presented in Figs. 9 to 12.



(a) Global diagram corresponding to Figs. 6 and 7.

(b) Localized enlargement with $\alpha \in [1, 2]$.**Fig. 8.** Integral of elevation angle corresponding to different alpha.**Fig. 9.** Enhanced obstacle avoidance trajectories corresponding to different fuel consumption.**Fig. 11.** Integral of elevation angle corresponding to different fuel consumption.**Fig. 10.** Elevation angles corresponding to different fuel consumption.

From Fig. 9, it can be observed that when the fuel consumption is optimal, the trajectory lands along the edge of the obstacle. As fuel consumption gradually increases, the trajectory shifts more toward the vertical upward direction. When the same amount of additional fuel is incrementally added, the trajectory changes more significantly in the early phase and less so in the later phase. Similarly, Figs. 10 and 11 demonstrate that as fuel consumption increases, the elevation angle changes more noticeably in the early phase and less so in the later phase. This is because upward flight must overcome the influence of gravitational acceleration, leading to an exponential increase in additional fuel consumption. Fig. 12 shows that the objective function also decreases with increasing fuel, and hence the trajectory has a better obstacle avoidance effect. Therefore, when the terminal time is fixed, applying Algorithm 2 with a slight increase in fuel consumption can significantly improve obstacle avoidance performance.

4.4. Trajectory analysis under fixed fuel consumption and varying terminal time

Within the feasible region described in Fig. 4, when fuel consumption is fixed, the terminal time has both an upper and a lower bound. In

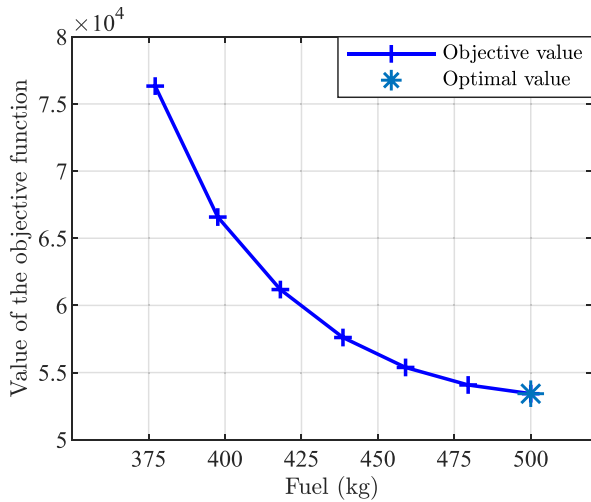


Fig. 12. Value of the objective function corresponding to different fuel consumption.

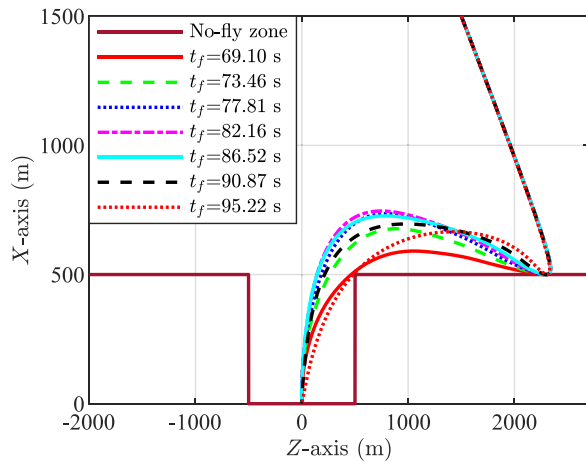


Fig. 13. Enhanced obstacle avoidance trajectories corresponding to different terminal times.

this subsection, the effects of terminal time on trajectory characteristics and elevation angles are investigated while keeping fuel consumption fixed. Specifically, when the fuel consumption is set to 400 kg and other parameter conditions are as listed in Tables 1 and 2, the resulting trajectories and elevation angles are shown in Figs. 13 to 16.

As illustrated in Fig. 13, the landing trajectory closely follows the edge of obstacles when the terminal time is set to its maximum or minimum values. However, when intermediate terminal time values are employed, the trajectory exhibits an upward displacement, with the degree of shift varying depending on the specific terminal time. Combined with the observations from Figs. 14 to 16, there exists a maximum value for the integral of the elevation angle and a minimum value for the value of the objective function. Notably, the elevation angle integral curve exhibits a single peak, indicating the existence of an optimal terminal time configuration for trajectory optimization. The terminal times corresponding to the maximum elevation angle integral and the minimum objective function value differ. This discrepancy arises because the objective function is integrated over time, whereas the elevation angle is integrated over altitude. This distinction underscores the rationale for adopting altitude-based integration of the elevation angle in this study, as it enables fair comparisons across different terminal times.

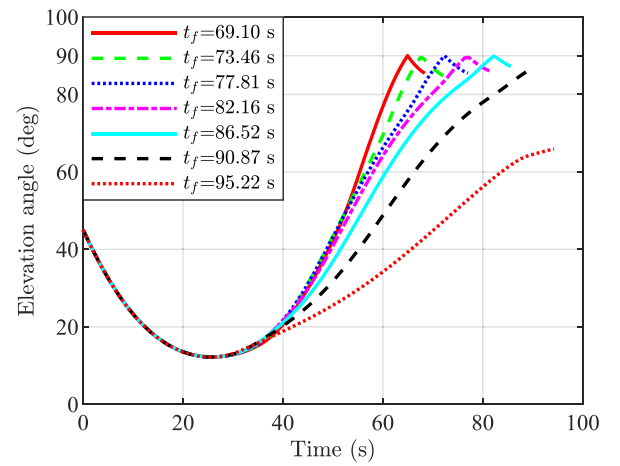


Fig. 14. Elevation angles corresponding to different terminal times.

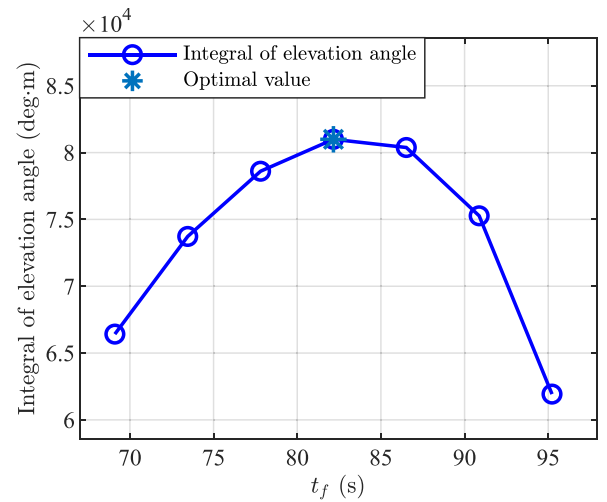


Fig. 15. Integral of elevation angle corresponding to different terminal times.

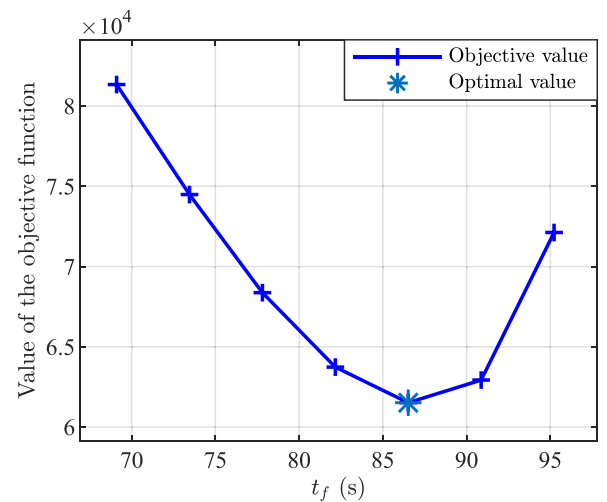


Fig. 16. Value of the objective function corresponding to different terminal times.

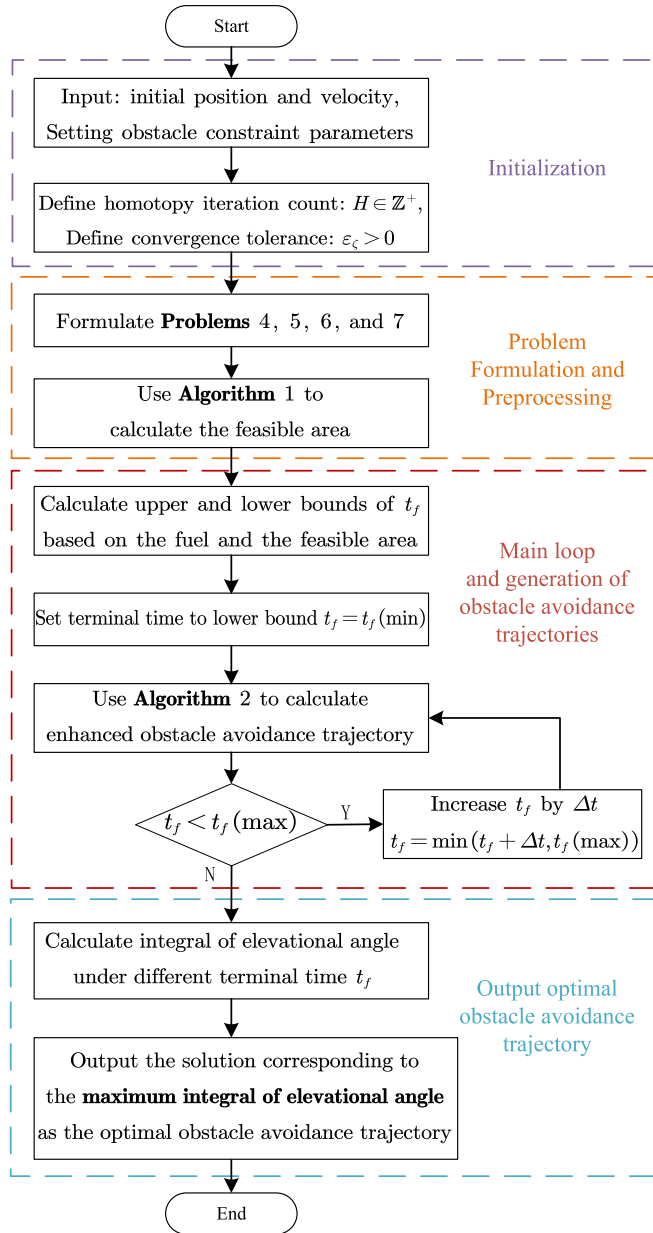


Fig. 17. Flowchart of powered descent obstacle avoidance guidance based on elevation angle maximization.

4.5. Elevation-angle-based optimal obstacle avoidance guidance

Based on the analysis in Sections 4.3 and 4.4, the following conclusion can be drawn: when the initial position, velocity, and fuel are given, and the terminal time is not fixed, there exists an optimal terminal time that maximizes the integral of the elevation angle, thereby achieving the best obstacle avoidance and field of view. This is referred to as elevation-angle-based optimal obstacle avoidance. Based on this, the algorithmic framework for obtaining the elevation-angle-based optimal obstacle avoidance trajectory is presented in Fig. 17.

5. Numerical experiments

In this section, the proposed Algorithm 1, Algorithm 2, and the elevation-angle-based optimal obstacle avoidance guidance algorithm outlined in Fig. 17 are applied to the Mars landing problem. First, the

Table 1
Simulation parameters for lander.

Parameters	Value	Units
m_{wet}	1905	kg
m_{dry}	1405	kg
T_{max}	13 258.17	N
T_{min}	4971.82	N
λ	5.09×10^{-4}	s/m
$\mathbf{g}_m$	$[-3.7114, 0, 0]^T$	m/s ²

Table 2
Simulation parameters for constraints.

Parameters	Value	Units
$\mathbf{r}_0$	$[1500, 0, 1500]^T$	m
$\mathbf{v}_0$	$[-75, 0, 70]^T$	m/s
$\mathbf{r}_f$	$\mathbf{0}_{3 \times 1}$	m
$\mathbf{v}_f$	$\mathbf{0}_{3 \times 1}$	m/s
t_f	80	s
α	1	–
W_ζ	10^3	–
l_1	100	m
l_2	500	m
l	500	m
h	500	m

simulation scenario, lander parameters, and obstacle constraints are presented. Subsequently, the effectiveness of proposed algorithms are validated. Finally, the robustness of the proposed elevation-angle-based optimal obstacle avoidance guidance algorithm is tested under various initial conditions.

The proposed algorithms were implemented in MATLAB R2023b using the CVX framework [45] and MOSEK solver [46]. Simulations were performed on a system featuring an Intel Core i9-13900HX (2.20 GHz) processor and 32 GB RAM.

5.1. Scenarios and parameters

The specific parameters for the lander and environment used in this study are provided in Table 1, and the initial and terminal positions and obstacle constraints are listed in Table 2. Some parameters are based on Ref. [20]. The coefficient α in Algorithm 2 is set to $\alpha = 1$. In the following simulations, if necessary to show the effectiveness of the algorithms under varying initial conditions, adjustments will be made to the initial position and velocity.

The convergence tolerance $\varepsilon_c = 10^{-6}$ follows standard practice in sequential convex programming. In all simulated cases, the final two-norm of the slack variables $\|\zeta(\cdot)\|_2$ reaches 10^{-10} , confirming that the relaxed solution tightly satisfies the original non-convex constraints. The number of homotopy iterations is set to $H = 3$ after a parametric study. With $H = 3$, the algorithm converges within four SCP iterations and $\|\zeta(\cdot)\|_2$ rapidly decreases to 10^{-11} , representing the minimum number of homotopy steps that yields stable and efficient convergence for the considered landing scenario. The corresponding convergence behavior is shown in Fig. 19.

5.2. Validation of Algorithm 1

In this section, the effectiveness of Algorithm 1 is validated through simulation examples. First, the thrust results obtained from the improved lossless convexification algorithm are compared with those from the conventional convex optimization algorithms [20,24] to demonstrate the effectiveness of the proposed improvements in addressing Taylor expansion errors. Second, fuel optimal trajectories under different obstacle constraints are provided to further validate the effectiveness of obstacle avoidance.

In the first validation, the initial position, velocity, and all other parameters are set according to the values provided in Tables 1 and 2.

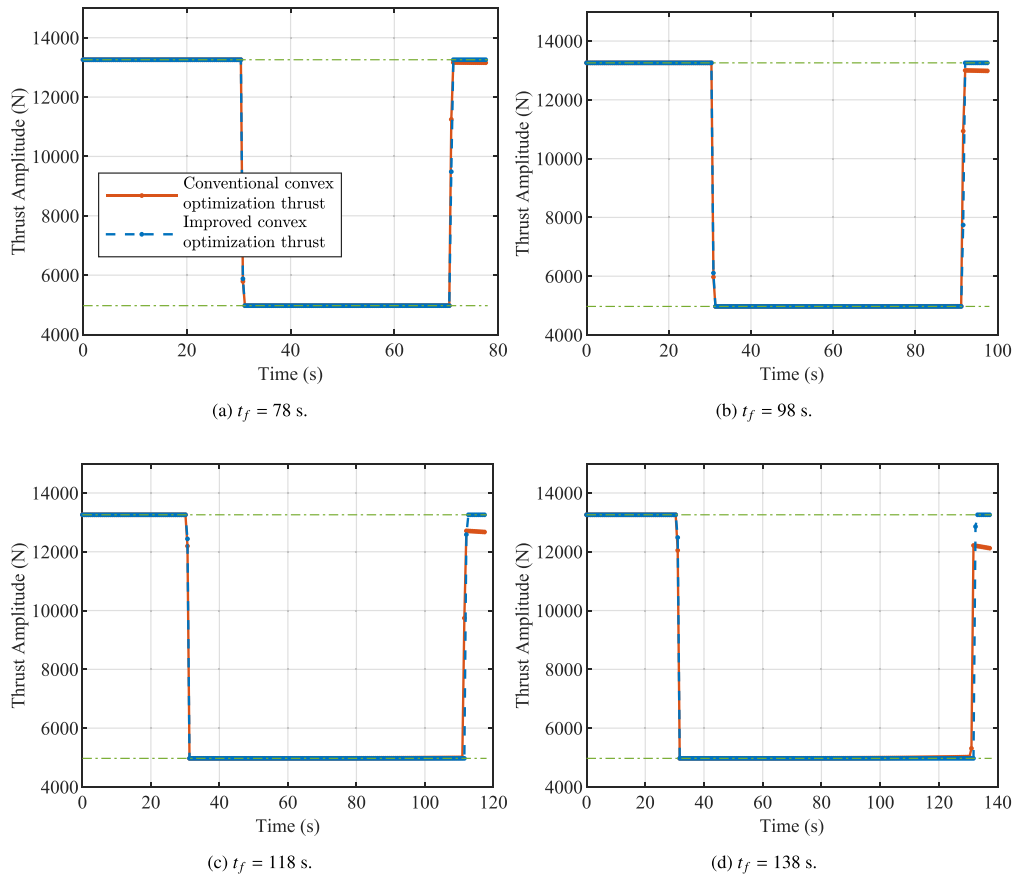


Fig. 18. Thrust comparison under varying terminal times for conventional and improved convex optimization methods.

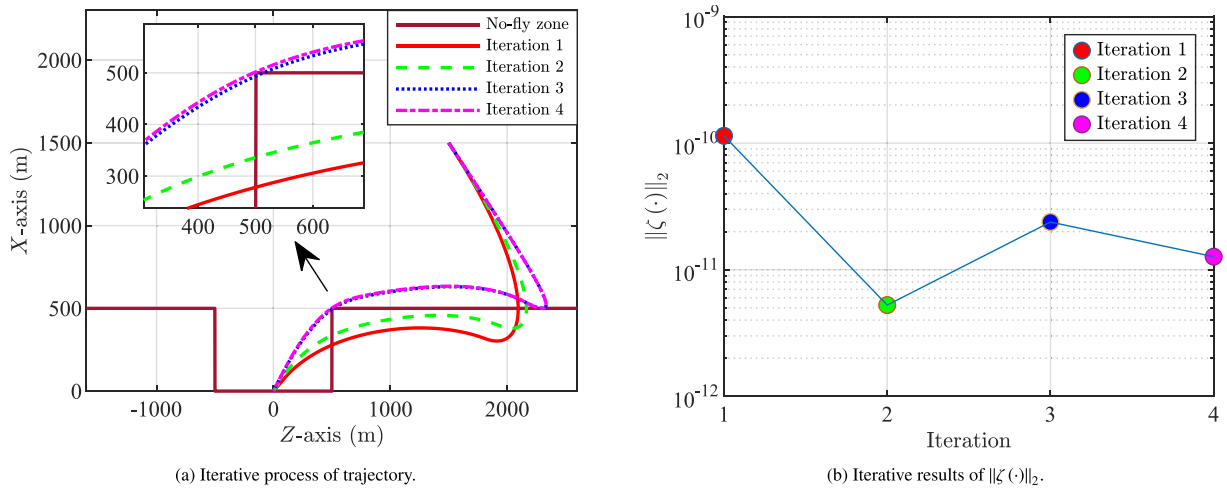


Fig. 19. Iterative process under stepwise constraint with $H=3$.

The obstacle is modeled using the conventional glide-slope constraint with an angle of $\gamma_{gs} = 10^\circ$. The results are presented in Fig. 18.

The subplots in Fig. 18 reveal that as the terminal time increases, the thrust profile obtained using the conventional convex optimization method gradually deviates from the maximum–minimum control profile, leading to a growing discrepancy between the solution and the optimal trajectory. In contrast, the improved convex optimization method effectively reduces the Taylor expansion errors that increases with terminal time, thereby yielding solutions that are closer to the true optimal trajectory.

Table 3 presents a comparison of the performance of the four methods for the scenario shown in Fig. 18(d). Both proposed algorithms require only 0.06 s of CPU time, which is comparable to the conventional convex method (0.03 s) and significantly lower than that of GPOPS (1.12 s), while achieving nearly identical fuel consumption (approximately 505.49 kg). Notably, the proposed methods are capable of handling step-shaped obstacles, a feature not supported by the conventional convex method. Furthermore, Algorithm 2 provides an additional advantage by improving the landing field of view.

Table 3
Performance metric of different methods.

Performance metric	Proposed method (Alg. 1)	Proposed method (Alg. 2)	Conventional convex method [20,24]	GPOPS software [47]
Average CPU time (s) (Scenario: Fig. 18 d)	0.06	0.06	0.03	1.12
Fuel consumption (kg) (Scenario: Fig. 18 d)	505.49	505.49	505.51	505.484
Step-shaped obstacle avoidance	Yes	Yes	No	Yes
Improve field of view	No	Yes	No	Yes

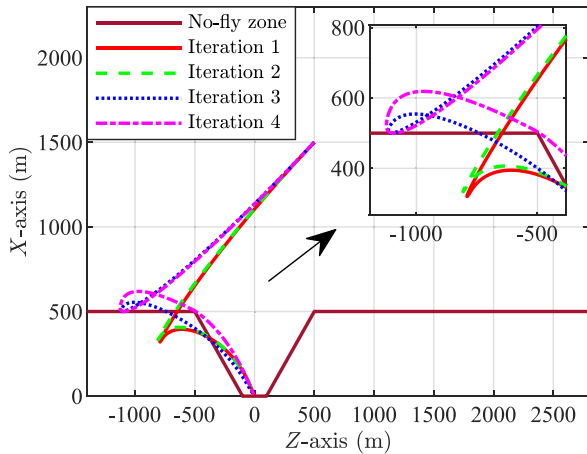


Fig. 20. Obstacle avoidance trajectory under relaxed glide-slope constraint.

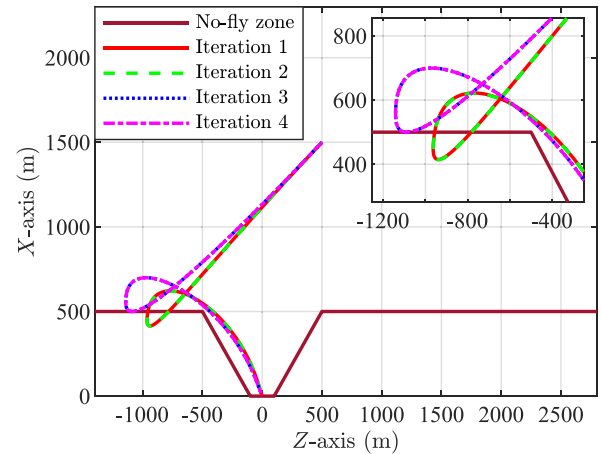


Fig. 21. Fuel optimal obstacle avoidance trajectory with $t_f = 100$ s.

In the second validation, when the obstacle is modeled as a stepwise constraint, as illustrated in Fig. 2, and the initial state is set according to Table 2. The iterative evolution of the trajectory and the slack variable norm $\|\zeta(\cdot)\|_2$ under Algorithm 1 is illustrated in Fig. 19. The figures show that the fuel optimal trajectory is achieved after four iterations, successfully accomplishing obstacle avoidance.

When the obstacle is configured as a relaxed glide-slope constraint, as shown in Fig. 2, and the initial state is set to $r_0 = [1500, 0, 500]^T$ m and $v_0 = [-75, 0, -100]^T$ m/s, with the remaining parameters given in Table 2, the iterative results for the fuel optimal obstacle avoidance trajectory are obtained using Algorithm 1, as illustrated in Fig. 20. The total fuel consumption in this case is 365.18 kg. From the figure, it can be seen that the fuel optimal obstacle avoidance trajectory is achieved after the same four iterations.

When the terminal time in Table 2 is set to $t_f = 100$ s, the resulting fuel optimal obstacle avoidance trajectory is shown in Fig. 21, with a fuel consumption of 409.12 kg. A comparison of Figs. 20 and 21 reveals that, despite the increase in fuel consumption, there is no significant improvement in obstacle avoidance performance.

5.3. Validation of Algorithm 2

In Section 4, simulation and analysis results under stepwise constraint were presented. This section sets the obstacle avoidance constraint as the relaxed glide-slope constraint and provides simulation results based on Algorithm 2 and elevation-angle-based optimal obstacle avoidance guidance in Fig. 17.

When the obstacle is configured as a relaxed glide-slope constraint, the initial state is set to $r_0 = [1500, 0, 500]^T$ m and $v_0 = [-75, 0, -100]^T$ m/s, consistent with Fig. 20, and the remaining parameters are as listed in Table 2. First, using Algorithm 1, the feasible region for fuel consumption and terminal time is obtained, as shown in Fig. 22.

From Fig. 22, it can be observed that the fuel optimal curve with respect to time exhibits a unimodal shape, and the overall structure

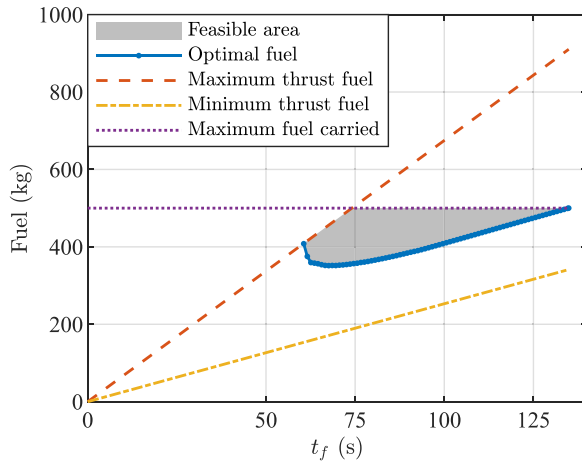
of the feasible region is similar to that in Fig. 4. However, unlike Fig. 4, the fuel consumption for the minimum landing time coincides with the maximum thrust fuel consumption curve, indicating that the thrust for the minimum landing time under these initial conditions and constraints is at full throttle. Additionally, the fuel consumption for the maximum landing time matches the total fuel carried.

When the fuel consumption is fixed at 365.18 kg, consistent with Fig. 20, the results shown in Figs. 23 to 26 are obtained using Algorithm 2. The resulting optimal obstacle avoidance trajectory demonstrates enhanced obstacle avoidance and provides a better terminal field of view. The resulting optimal obstacle avoidance trajectory exhibits improved avoidance performance and offers a better terminal field of view. However, as observed in the figures, although some improvement is achieved, the enhancement in obstacle avoidance is not particularly significant.

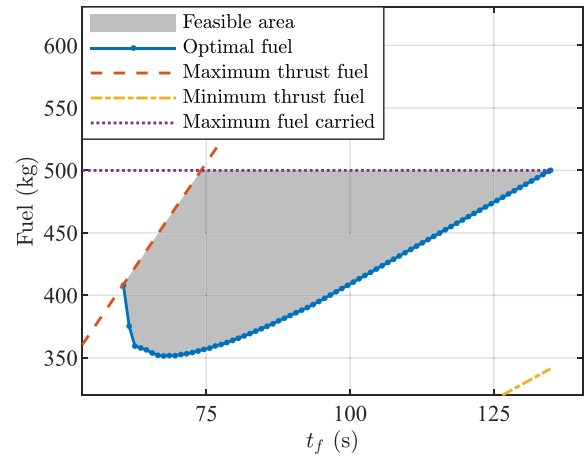
When maintaining a fixed fuel consumption of 409.12 kg, consistent with Fig. 21, by employing Algorithm 2, a series of obstacle avoidance trajectories are obtained as shown in Figs. 27 and 28. The integral of elevation angle and value of objective function are presented in Figs. 29 and 30. The results show a significant improvement in obstacle avoidance in Fig. 27 compared to the original trajectory in Fig. 21, while maintaining the same fuel consumption and providing a better terminal field of view.

5.4. Validation of elevation-angle-based optimal obstacle avoidance guidance

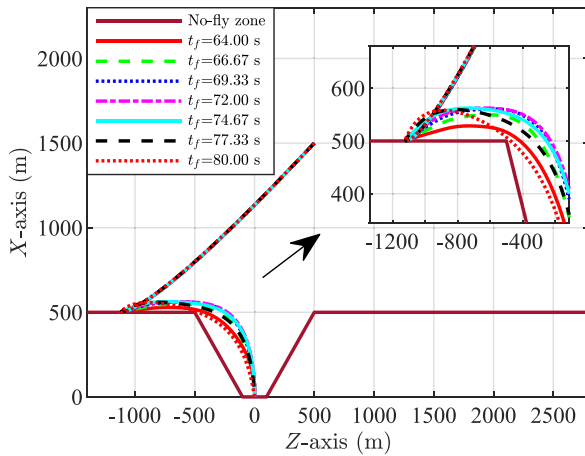
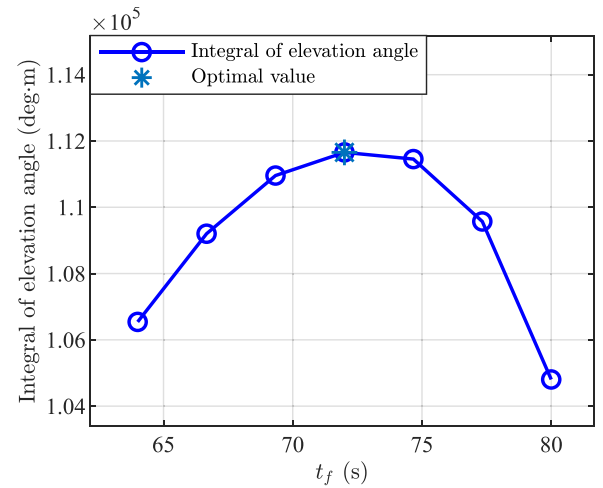
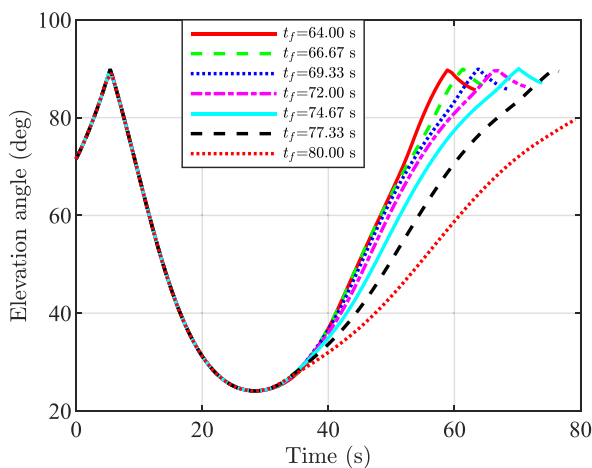
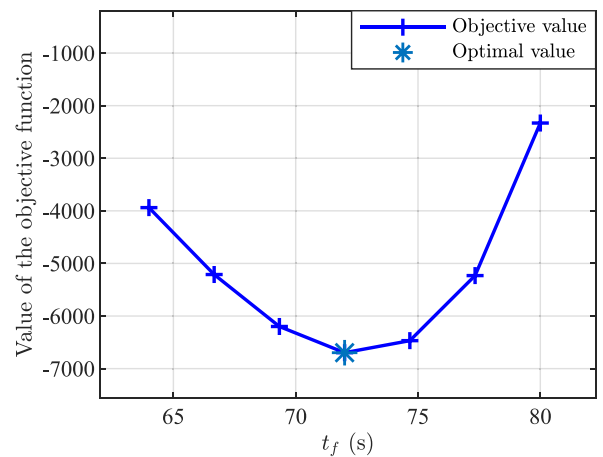
In this section, to maintain consistency with Fig. 4, stepwise constraint is applied, and the initial conditions are set as shown in Table 1 and 2. Based on the analyses in Section 4.4 and Section 4.5, when fuel consumption is fixed, utilizing the guidance flowchart in Fig. 17 yields an optimal obstacle avoidance trajectory. When the mass varies, the results are shown in Figs. 31 and 32. Fig. 31 illustrates the



(a) Global diagram of feasible area.



(b) Localized enlargement of feasible region.

Fig. 22. Illustration of feasible area for fuel consumption and terminal time t_f .**Fig. 23.** Enhanced obstacle avoidance trajectories corresponding to different terminal times.**Fig. 25.** Integral of elevation angle corresponding to different terminal times.**Fig. 24.** Elevation angles corresponding to different terminal times.**Fig. 26.** Value of the objective function corresponding to different terminal times.

relationship between the elevation angle integral and the terminal time under different fuel consumption, while Fig. 32 depicts the relationship between the objective function value and the terminal time under different fuel consumption.

From Fig. 31, it can be observed that when the terminal time t_f remains constant, the integral of the elevation angle increases with higher fuel consumption. When fuel consumption is fixed, the integral

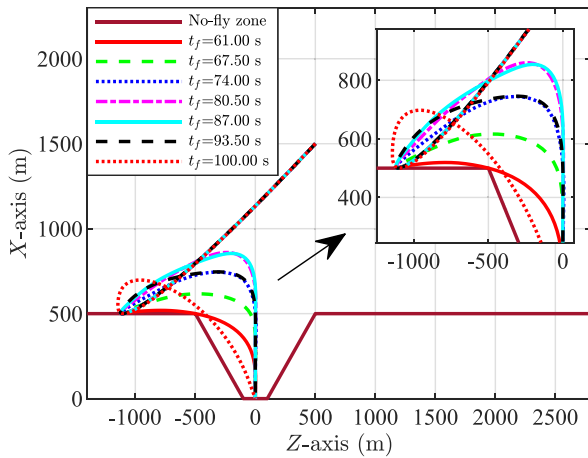


Fig. 27. Enhanced obstacle avoidance trajectories corresponding to different terminal times.

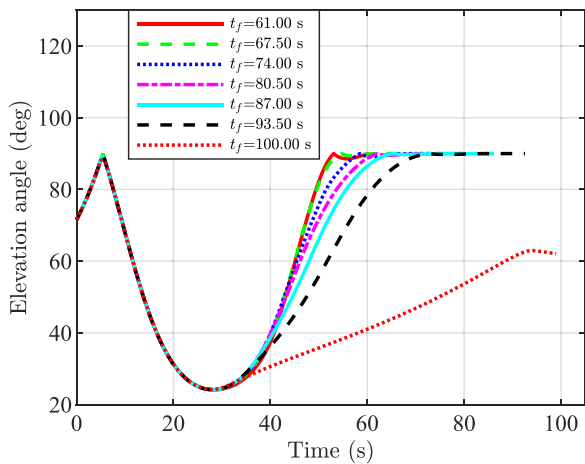


Fig. 28. Elevation angles corresponding to different terminal times.

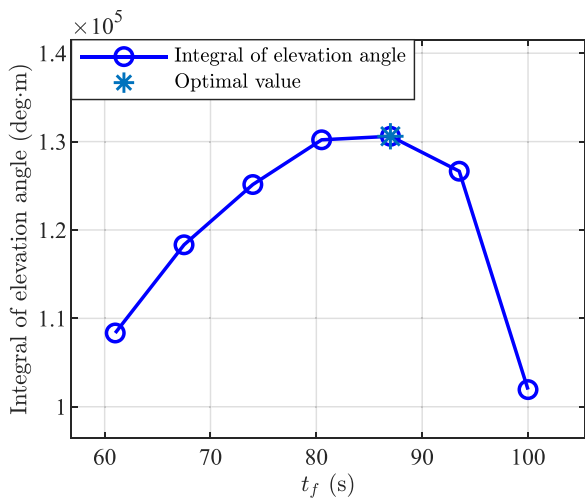


Fig. 29. Integral of elevation angle corresponding to different terminal times.

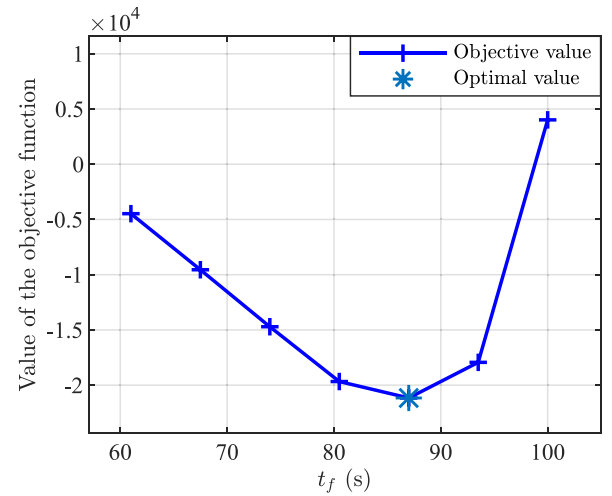


Fig. 30. Value of the objective function corresponding to different terminal times.

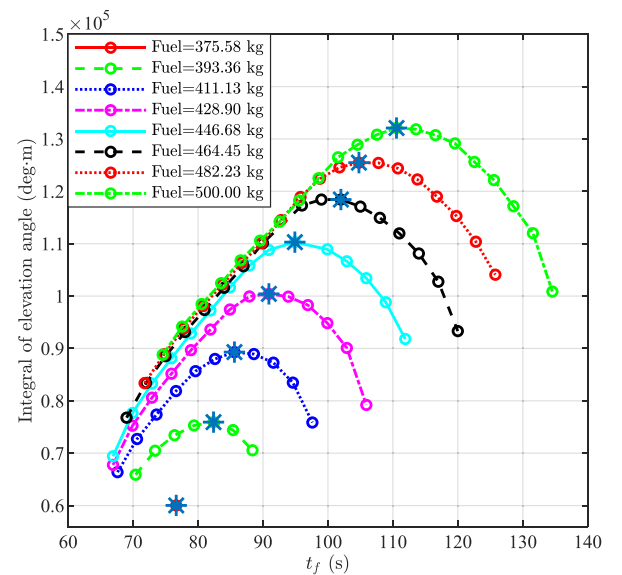


Fig. 31. Integral of elevation angle corresponding to different terminal times and fuel consumption.

of the elevation angle as a function of time exhibits a unimodal curve, with a distinct maximum value. Additionally, for the same integral of the elevation angle, there exists a range of fuel and terminal time pairs, meaning that both lower and higher fuel consumption can yield the same effect. In such cases, selecting the lower fuel consumption is appropriate. As shown in Fig. 31, the variations in the elevation angle integral caused by changes in terminal time are less pronounced at lower fuel consumption levels, but they become more significant as fuel consumption increases. This also explains the differing degrees of trajectory variation observed in Figs. 20 and 21.

From Fig. 32, it can be observed that the value of objective function exhibits a trend similar to that of the integral of elevation angle. When the terminal time t_f remains constant, the objective function value decreases as fuel consumption increases. When fuel consumption is fixed, the objective function value as a function of time follows a unimodal curve, with a distinct minimum value. Additionally, for the same objective function value, there also exists a range of fuel and terminal time pairs.

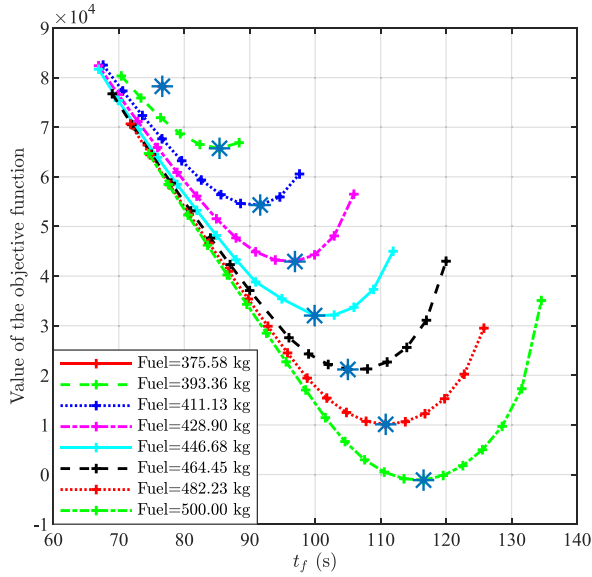


Fig. 32. Objective value corresponding to different terminal times and fuel consumption.

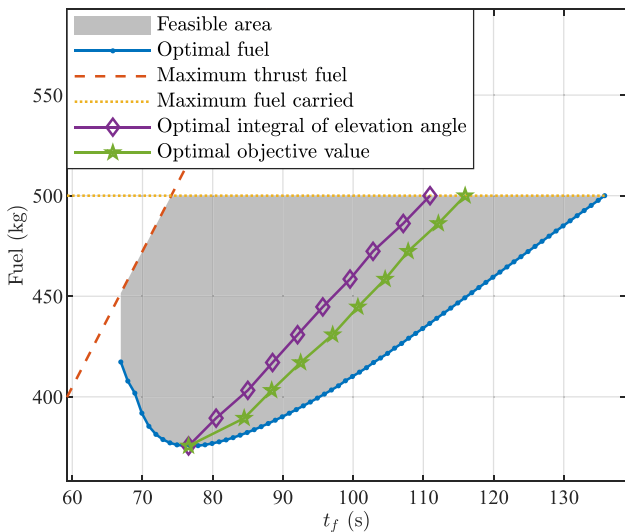


Fig. 33. Optimal integral of elevation angle and optimal objective value in feasible area.

However, since the objective function is an axis-weighted position integral evaluated over time, it is not ideal for accurately comparing trajectories with different terminal times. In contrast, integrating the elevation angle over altitude provides a fairer basis for evaluating field-of-view performance, making it more suitable for selecting the optimal trajectory.

Therefore, for a given fuel consumption, there exists an optimal terminal time that maximizes the elevation angle integral. By plotting these data points within the feasible region, an optimal curve can be derived, as shown in Fig. 33. The figure shows that the overall trends of the two optimal curves are consistent. Consequently, for any given fuel consumption within the feasible region, both the solution that maximizes the elevation angle and the one that minimizes the objective function can be directly determined from these curves.

Table 4

Initial conditions for two-dimensional cases.

Cases	Initial position r_0 (m)	Initial velocity v_0 (m/s)	Terminal time t_f (s)
Case 1	$[1500, 0, -2000]^T$	$[-75, 0, -10]^T$	80
Case 2	$[1500, 0, -1000]^T$	$[-75, 0, -50]^T$	80
Case 3	$[1500, 0, 0]^T$	$[-75, 0, -40]^T$	60
Case 4	$[1500, 0, 1000]^T$	$[-75, 0, -10]^T$	60
Case 5	$[1500, 0, 2000]^T$	$[-75, 0, -30]^T$	80

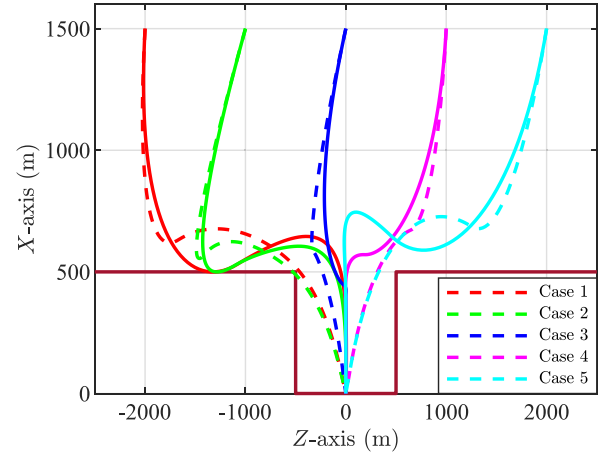


Fig. 34. Obstacle avoidance trajectories under different initial conditions.

Table 5

Initial conditions for 3-DoF cases.

Cases	Initial position r_0 (m)	Initial velocity v_0 (m/s)	Terminal time t_f (s)
Case 1	$[1500, 2000, 500]^T$	$[-75, -20, 10]^T$	80
Case 2	$[1500, 500, 2000]^T$	$[-75, 0, -30]^T$	80
Case 3	$[1500, 1000, 2000]^T$	$[-75, 20, 20]^T$	80
Case 4	$[1500, 2000, 1000]^T$	$[-75, -10, 10]^T$	80
Case 5	$[1500, 2000, 2000]^T$	$[-75, 10, -10]^T$	80

5.5. Robustness of algorithms and 3-DoF cases

5.5.1. Robustness under different initial conditions

In this section, the robustness of Algorithm 1 and the elevation-angle-based optimal obstacle avoidance guidance method illustrated in Fig. 17 is evaluated under different initial conditions. The initial conditions are set as shown in Table 4. By applying Algorithm 1 and flowchart in Fig. 17, the fuel-optimal trajectories and the elevation-angle-optimal trajectories are obtained, yielding the results illustrated in Fig. 34, where the dashed line indicates results of Algorithm 1 and the solid line indicates results of Algorithm in Fig. 17. The fuel consumption of the optimal obstacle avoidance guidance is based on the fuel consumption obtained from Algorithm 1 and remains the same.

From Fig. 34, it can be seen that all trajectories successfully avoid obstacles. The elevation angle-based method, however, offers a better terminal field of view while maintaining equivalent fuel consumption. Additionally, this method achieves significantly reduced lateral velocity during final descent, thereby enhancing landing safety.

5.5.2. 3-DoF cases

Unlike the analytical guidance, the optimization method can be directly applied to 3-DoF cases, with the initial conditions as in Table 5 and the corresponding results as shown in Fig. 35, where the dashed line denotes the fuel optimal and the solid line denotes the elevation-angle-based optimal obstacle avoidance. The fuel consumption of the optimal obstacle avoidance guidance is based on the fuel consumption obtained from Algorithm 1 and remains the same.

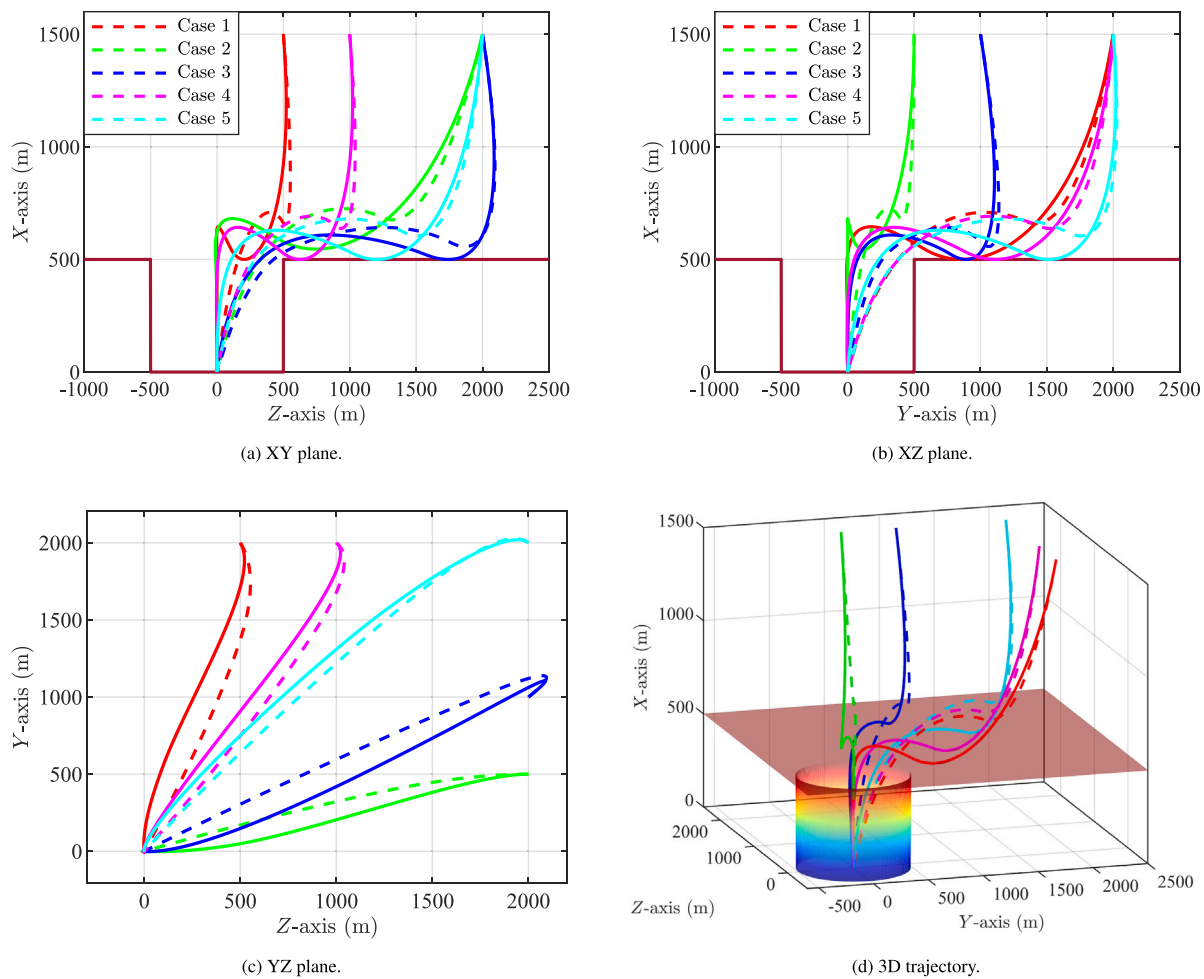


Fig. 35. Obstacle avoidance trajectories under different 3-DoF cases and their projections onto XY, XZ, and YZ planes.

As shown in Fig. 35, under different conditions, both Algorithm 1 and the optimal obstacle avoidance guidance in Fig. 34 successfully achieve effective obstacle avoidance. However, given the same fuel consumption, the optimal obstacle avoidance guidance provides superior terminal field of view and improved obstacle avoidance performance in comparison.

6. Conclusions

This paper introduced an elevation-angle-based convex optimization framework for obstacle avoidance guidance during the Mars powered descent phase. The proposed method integrates relaxed glide-slope and stepwise obstacle constraints while enhancing the terminal field of view to enable near-vertical landing. Scenario analyses reveal that under fixed fuel and terminal time, the trajectory shape can be modulated via objective function coefficients, with feasible ranges determined by the elevation-angle integral. When terminal time is fixed, increased fuel consumption improves the elevation angle, though with diminishing returns. Under fixed fuel consumption, the algorithm achieves superior obstacle avoidance and enhanced terminal visibility without exceeding the fuel budget. Based on these findings, a guidance framework utilizing the optimal elevation-angle integral is constructed, which improves obstacle avoidance while preserving fuel efficiency. The resulting near-vertical landing configuration also substantially mitigates lateral tipping risk, thereby enhancing landing safety. Future work will extend this framework to 6-DoF powered descent scenarios, where the near-vertical landing property is expected to further reduce lander tipping risk.

CRediT authorship contribution statement

Duozhi Gao: Writing – review & editing, Writing – original draft, Visualization, Investigation, Conceptualization. **Youmin Gong:** Validation, Supervision. **Yanning Guo:** Validation, Supervision, Funding acquisition. **Yizheng Xiao:** Validation. **Edoardo Fadda:** Validation, Supervision. **Paolo Brandimarte:** Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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